TRADE THAT TIES: TRADE COMMUNITIES OVER TWO CENTURIES (1833–2025)

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INTRODUCTION

Caveat emptor

What do we call trade communities?

Why are we interested?

Trade communities play a role in world trade

Gopinath et al. (2025), Bonadio et al. (2026), Bhagwati (1991…), Anderson and Norheim (1993),

We look for trade communities with network science

Fagiolo et al. (2010) De Benedictis & Tajoli (2011), Piccardi & Tajoli (2012), Iapadre & Tajoli (2014)

Trade communities have economic consequences.

Lesger (2006), Greif (1993), Miller (2012), Rauch (1999, 2001), Rauch & Trindade (2002), Combes, Lafourcade & Mayer (2005), Chaney (2014)

What are the composition, determinants and effects of trade communities over two centuries ?

Outline

1. Data
2. Identifying trade communities
3. Finding the determinants of trade communities
4. Finding the effect of trade communities

I. DATA

DATA (1)

Well known IMF data from 1948

A database of bilateral world trade 1833-1938 (Girard et al. (2016))

RICardo project since 2004. 690,000 trade flows

https://ricardo.medialab.sciences-po.fr/

RICardo

Trade Data Glossary About

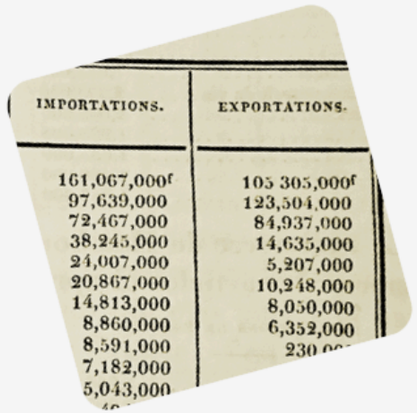
FR | EN

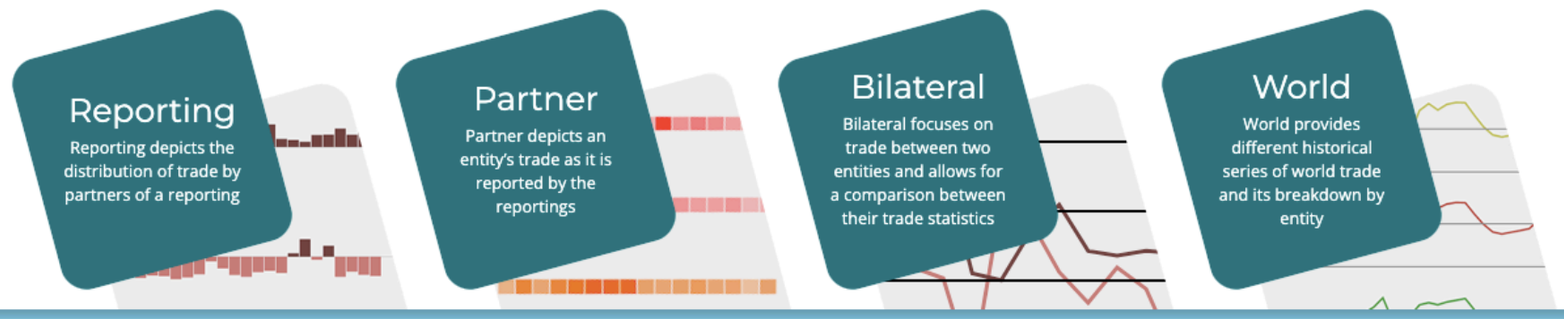
The RICardo Project

Trade between Nations from c. 1800 to 1938

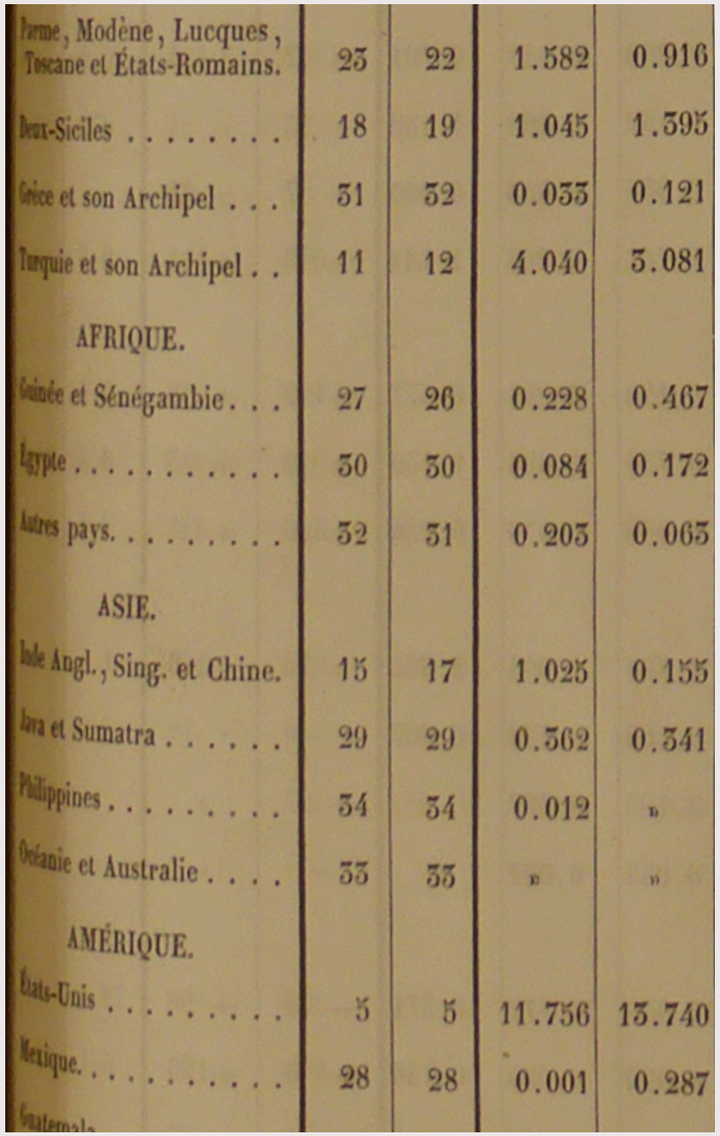
RICardo (Research on International Commerce) is a project dedicated to trade between nations over a period spanning the beginning of the Industrial Revolution to the eve of the Second World War.

It combines a historical trade database covering all of the world's countries and a website which invites to an exploration of the history of international trade through data visualizations.





DATA (2)



Parme, Modène, Lucques, Toscane et États-Romains.	23	22	1.582	0.916
Deux-Siciles	18	19	1.045	1.395
Grèce et son Archipel . . .	31	32	0.033	0.121
Turquie et son Archipel . .	11	12	4.040	3.081
AFRIQUE.				
Guinée et Sénégambie . . .	27	26	0.228	0.467
Égypte	30	30	0.084	0.172
Autres pays.	32	31	0.203	0.063
ASIE.				
Inde Angl., Sing. et Chine.	15	17	1.025	0.155
Java et Sumatra	29	29	0.362	0.341
Philippines	34	34	0.012	»
Océanie et Australie	33	33	»	»
AMÉRIQUE.				
États-Unis	5	5	11.756	13.740
Mexique.	28	28	0.001	0.287

Issue with the sources: Heterogeneity of trade entities in trade archives

Tableau général du commerce avec les pays étrangers (Belgique, 1850)\

DATA (3) : TRYING TO BE FAITHFUL TO THE SOURCES

Source: RICardo database

Flow (1000 £)	year	expimp	reporting	partner
1427004.4	1850	Exp	Belgium	United Kingdom
1561.9	1850	Exp	Belgium	Africa (other)
55802.4	1850	Exp	Belgium	Austria-Hungary (Austrian Empire)
103170.1	1850	Exp	Belgium	Brazil
80560.1	1850	Exp	Belgium	Chile & Peru
55841.5	1850	Exp	Belgium	Cuba & Puerto Rico
11402.6	1850	Exp	Belgium	Denmark
42174	1850	Exp	Belgium	Two Sicilies
5584.1	1850	Exp	Belgium	Egypt Arab Republic (Egypt)
64002.9	1850	Exp	Belgium	Spain
377691.6	1850	Exp	Belgium	United States of America
2608227.6	1850	Exp	Belgium	France
62245.7	1850	Exp	Belgium	Luxemburg
3983.1	1850	Exp	Belgium	Greece & Ionian Is.
8512.8	1850	Exp	Belgium	Guatemala
31981.9	1850	Exp	Belgium	Gambia & Guinea (French Guinea) (Rivieres du Sud) & Senegal
2967.7	1850	Exp	Belgium	Haiti & Venezuela
220944.8	1850	Exp	Belgium	Hanover & Oldenburg
11207.3	1850	Exp	Belgium	India & People's Republic of China (China) & Singapore
7302.3	1850	Exp	Belgium	Java & Sumatra
6404.2	1850	Exp	Belgium	Mecklenburg Schwerin
26475.9	1850	Exp	Belgium	Mexico
6365.1	1850	Exp	Belgium	Australia & Oceania
31513.3	1850	Exp	Belgium	Lucca & Modena & Papal States & Parma & Tuscany
1221796.4	1850	Exp	Belgium	Netherlands
117.1	1850	Exp	Belgium	Philippines
2303.9	1850	Exp	Belgium	Portugal
0	1850	Exp	Belgium	British America
1261080.6	1850	Exp	Belgium	Prussia
13862.7	1850	Exp	Belgium	Argentina (La Plata)
95984.8	1850	Exp	Belgium	Russia (USSR)
151474.9	1850	Exp	Belgium	Piedmont & Sardinia
1757.2	1850	Exp	Belgium	Bavaria & Frankfurt & Saxony
29248.4	1850	Exp	Belgium	Norway & Sweden
30146.5	1850	Exp	Belgium	Switzerland
8201788.6	1850	Exp	Belgium	World as reported
126639.1	1850	Exp	Belgium	Archipelago & Turkey (Ottoman Empire)
14448.4	1850	Exp	Belgium	Hanse Towns

Source: TRADEHIST (Fouquin & Hugit 2016)

ISO_o	ISO_d	year	Flow Exp (1000 £)
BEL	2SICIL	1850	42174
BEL	ARG	1850	13862.8
BEL	AUTHUN	1850	55802.4
BEL	BRA	1850	103170.1
BEL	CHE	1850	30146.6
BEL	CHL	1850	5583.7
BEL	CUB	1850	55841.5
BEL	DNK	1850	11402.6
BEL	EGY	1850	5584.1
BEL	ESP	1850	7891.5
BEL	FRA	1850	4113120
BEL	GBR	1850	1520194
BEL	GRC	1850	3983.1
BEL	GTM	1850	8512.9
BEL	IDN	1850	7302.4
BEL	LUX	1850	62245.7
BEL	MAR	1850	2755.2
BEL	MECKL	1850	6404.2
BEL	MEX	1850	26475.9
BEL	NLD	1850	1929164.3
BEL	OTTO	1850	126639.2
BEL	PHL	1850	117.2
BEL	PRT	1850	2303.9
BEL	PRUS	1850	1261081
BEL	SEN	1850	31981.9
BEL	SWENOR	1850	29248.4
BEL	USA	1850	377691.6
BEL	USSR	1850	95984.9
BEL	ZOLL	1850	1325083.8

DATA (4) STANDARDIZATION OF NAMES (INCLUDING TRANSLATION)

From c. 10,000 names to c. 2.500 names (COW : 1,200)

COW code	COW name	ISO code	ISO name
7501	Andaman and Nicobar Is.	-	-
242	Anhalt Cothen	-	-
5813	Anjouan	-	-
8152	Annam	-	-
7588	Baluchistan (Kalat)	-	-
80	Belize (British Honduras)	BLZ	Belize
434	Benin (Dahomey)	BEN	Benin
42	Dominican Republic (Santo Domingo)	DOM	Dominican Republic
404	Guinea-Bissau (Portuguese Guinea)	GNB	Guinea-Bissau

DATA (5)

There are very trading actors in the sources

RIC type	All entities	Reporting	Partner	Examples
Polity	530 (22.7%)	286	483	Bulgaria, Gambia, Singapore, Queensland…
Locality	707 (30.2%)	120	571	Asian Russia, Bahia, France (Atlantic Coast)…
Group	860 (36.8%)	37	797	Altona & Holstein, Trinidad & United Kingdom…
Colonial area	130 (5.6%)	0	120	British Africa, Portuguese Asia, Spanish West Indies…
Geographical area	112 (4.8%)	4	94	Central America, Eastern Africa, Persian Gulf…
Total	**2339**	**447**	**2065**	

To identify "British Africa" we need to know what are the British African colonies in a given year.

DATA (6)

A database of geopolitical entities of the world since 1816 (Dedinger & Girard (2021)

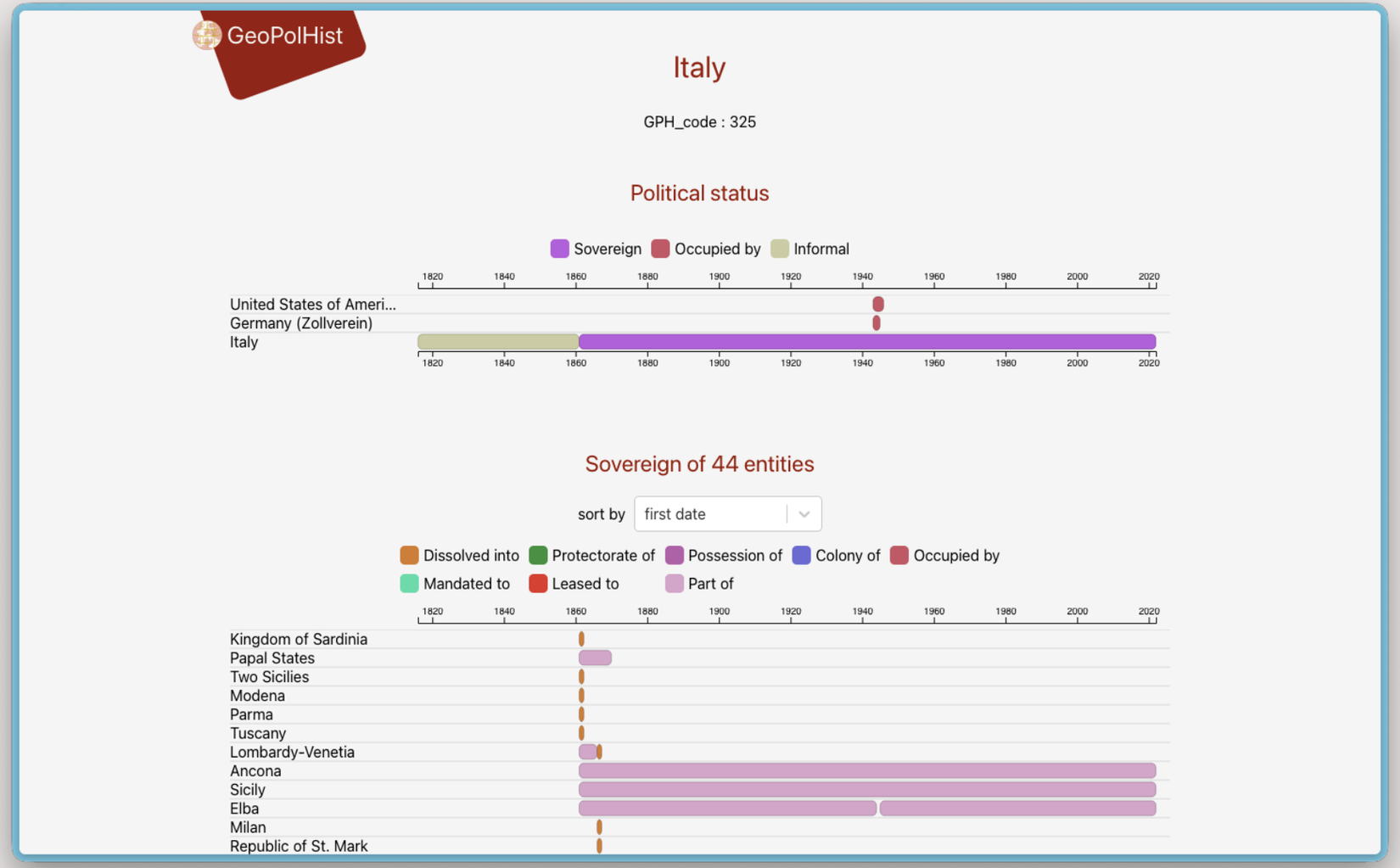
GeoPolHist

https://medialab.github.io/GeoPolHist/#/GeoPolHist/

DATA (7)
The GeoPolHist dataset

GPH code	GPH name	Start year	End year	GPH Status	GPH sovereign code
325	Italy	1816	1861	Informal	
325	Italy	1861	2022	Sovereign	
325	Italy	1943	1946	Occupied by	2
325	Italy	1943	1945	Occupied by	255
326	Trieste	1816	1918	Part of	300
326	Trieste	1918	1943	Part of	325
326	Trieste	1943	1947	Occupied by	2
326	Trieste	1947	1954	Mandated to	1
326	Trieste	1954	2022	Part of	325
327	Papal States	1816	1861	Sovereign	
327	Papal States	1861	1870	Part of	325
327	Papal States	1870	1870	Dissolved into	328
328	Vatican City	1816	1870	Part of	327
328	Vatican City	1870	1929	Part of	325
328	Vatican City	1929	1964	Sovereign (unrecognized)	
328	Vatican City	1964	2022	Sovereign	
329	Two Sicilies	1816	1860	Sovereign	
329	Two Sicilies	1860	1861	Part of	324
329	Two Sicilies	1861	1861	Dissolved into	325

GeoPolHist: Italy (325)



DATA (8)

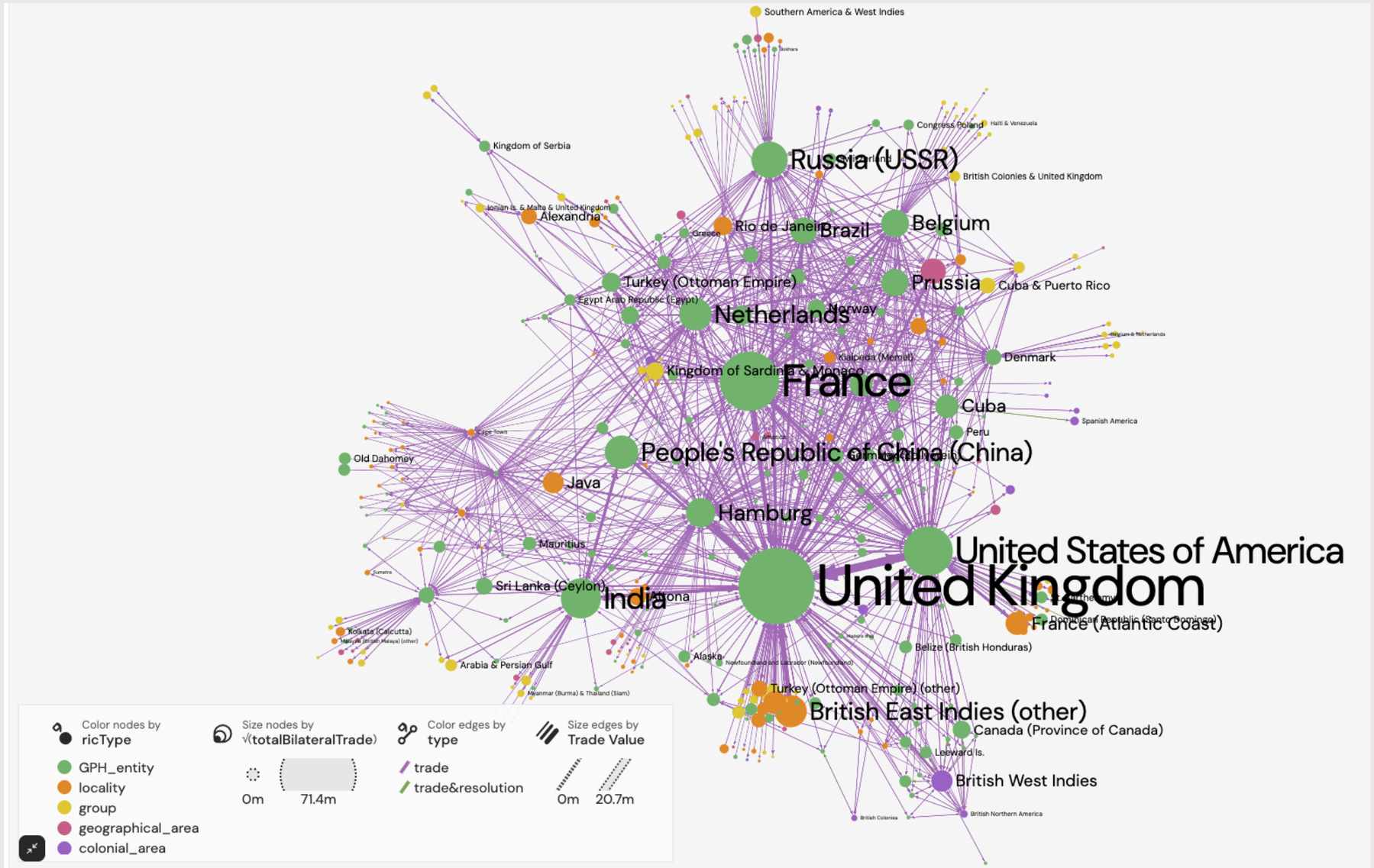
To build coherent bilateral trade matrices, one needs to reduce trading actors heterogeneity thanks to GeoPolHist data

1. identify non-polities, groups… informal entities.
2. aggregate non-polities entities (localities, part-of…) to their polities.
3. split trade of groups, geographical and colonial areas

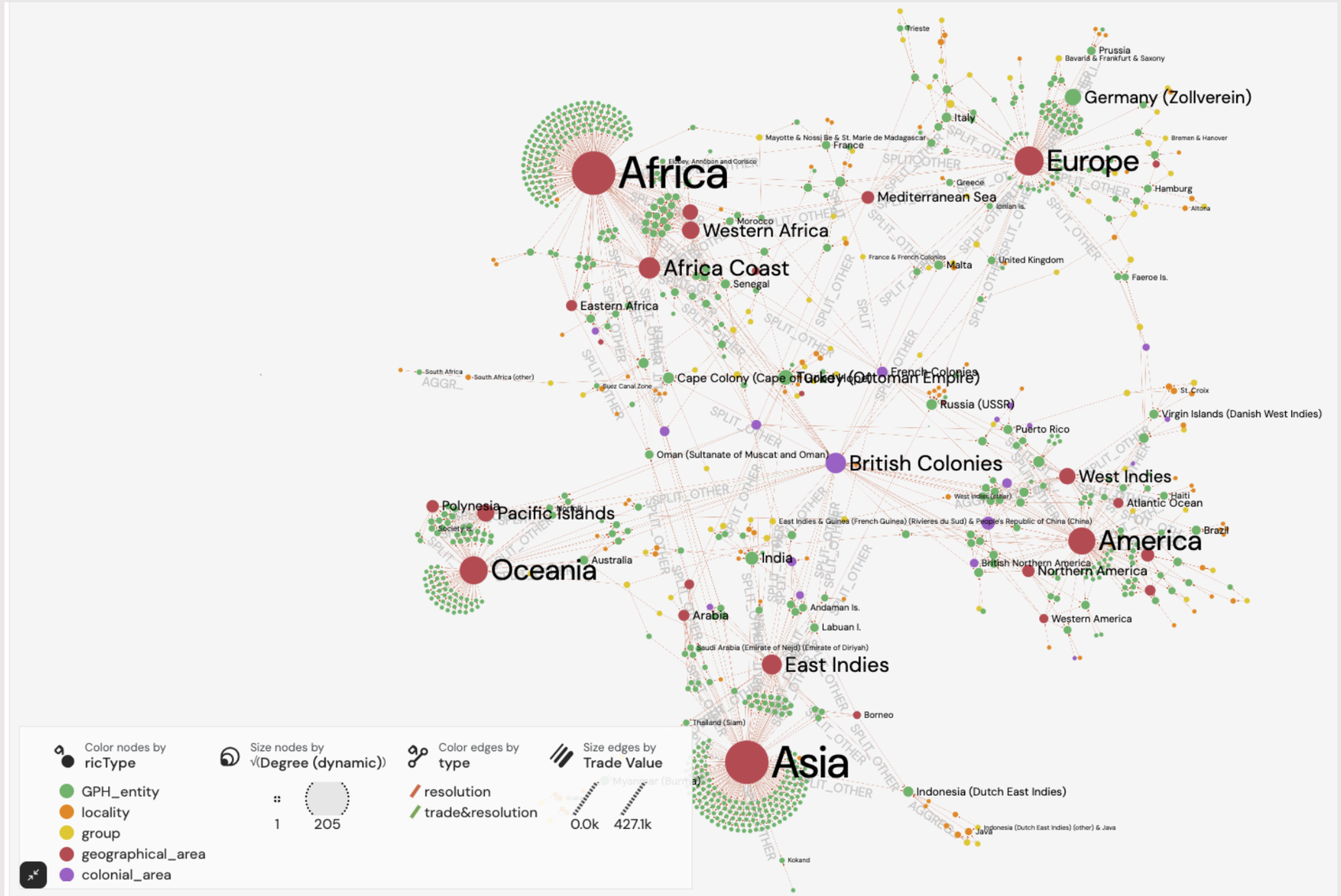
DATA (9)

Method: Multilayer networks combining trade and politics

The 1850 trade network



The 1850 resolution network



DATA (10)

Harmonization process

1. Polities resolution
2. Trade reporters aggregation/split
3. Trade partners aggregation
4. Trade partners splits with year ratio method
5. Trade reporters & partners splits with gravity model method

!\[===/assets/trade data model.png)

TRADE PARTNER SPLITS WITH YEAR RATIO METHOD

Difficult task: how to decide the ratios to split one trade value into many?

We look into **adjacent years** (+/- 10-years window) networks\

TRADE PARTNER/REPORTER SPLITS WITH GRAVITY MODEL METHOD

We try to impute flows we couldn't split with the adjacent years method by using a ppml gravity model (Anderson et Van Wincoop (2003), Santos Silva & Tenreyro (2006)).

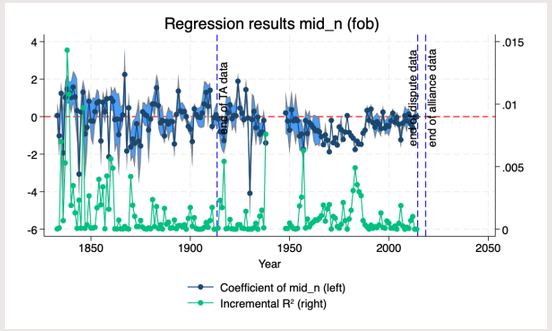
We do this cross-section. Very far from a balanced panel

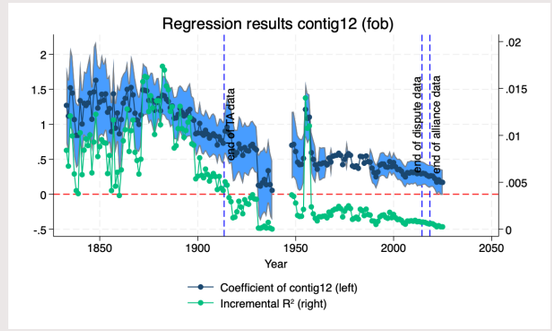
DATA (11)

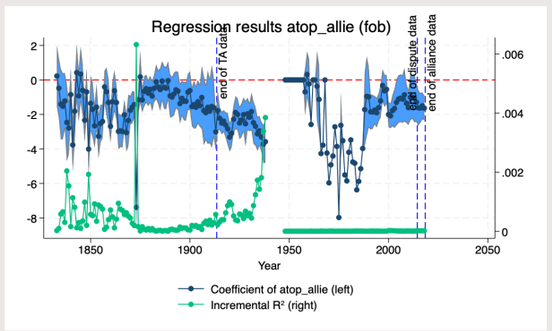
Control variables

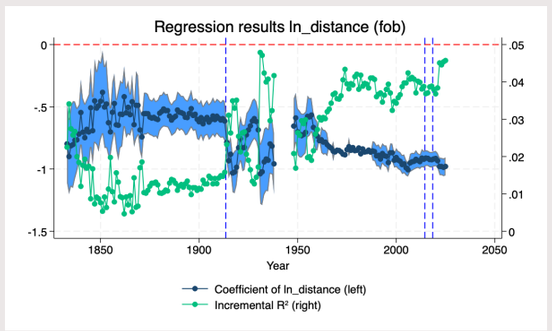
- Country x direction fixed effects
- distance (Centroids from Wikidata, not time varying)
- contiguity (Correlates of War + our own work)
- common empire (GeoPolHist up to 2022)
- alliance (inherited from sovereign) (Alliance Treaty Obligations and Provisions (ATOP) up to 2018)
- conflict (inherited from sovereign) (Militarized Interstate Disputes (MID) up to 2014)
- Trade agreements (Pahre (2008)'s data up to 1913)

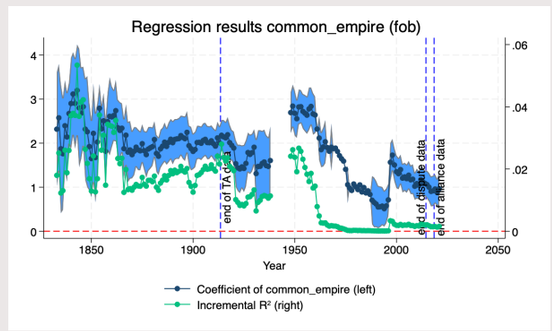

Regression results TA (fob)
end of TA data
Year
Coefficient of TA (left)
Incremental R² (right)

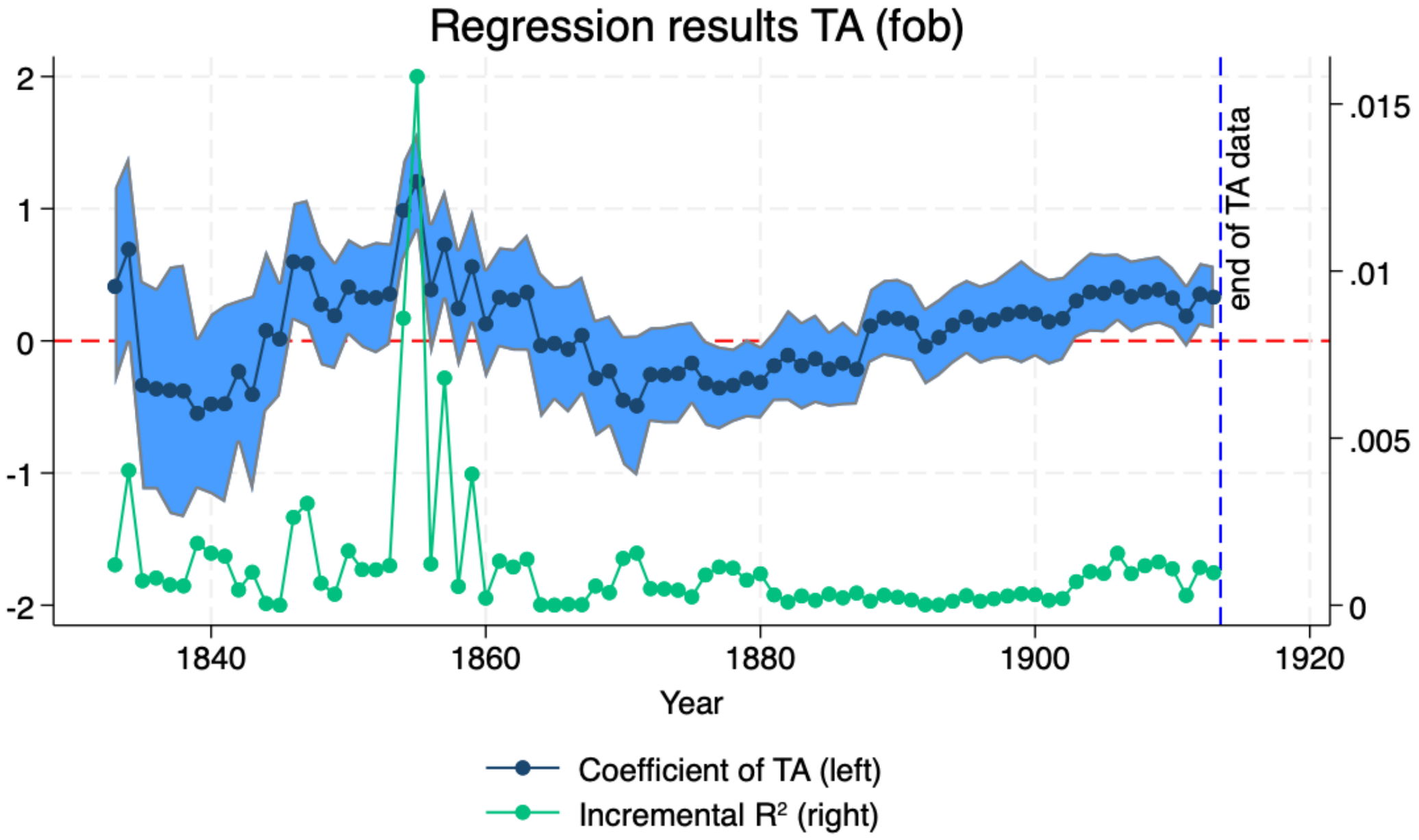

Regression results mid_n (fob)
end of TA data
end of dispute data
end of alliance data
Year
Coefficient of mid_n (left)
Incremental R² (right)

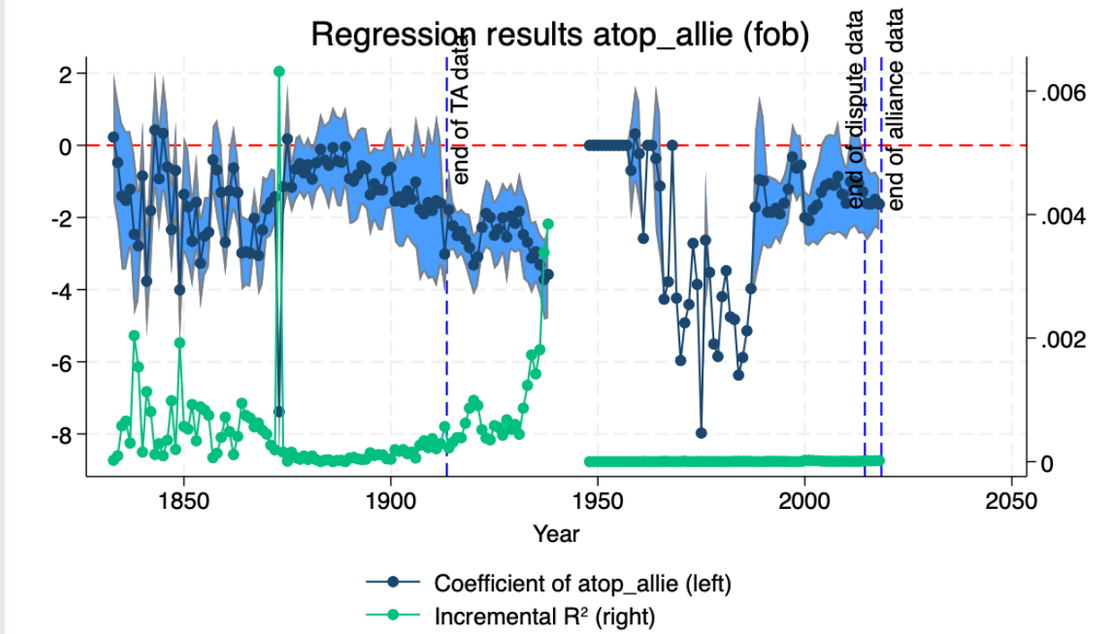

Regression results contig12 (fob)
end of TA data
end of dispute data
end of alliance data
Year
Coefficient of contig12 (left)
Incremental R² (right)

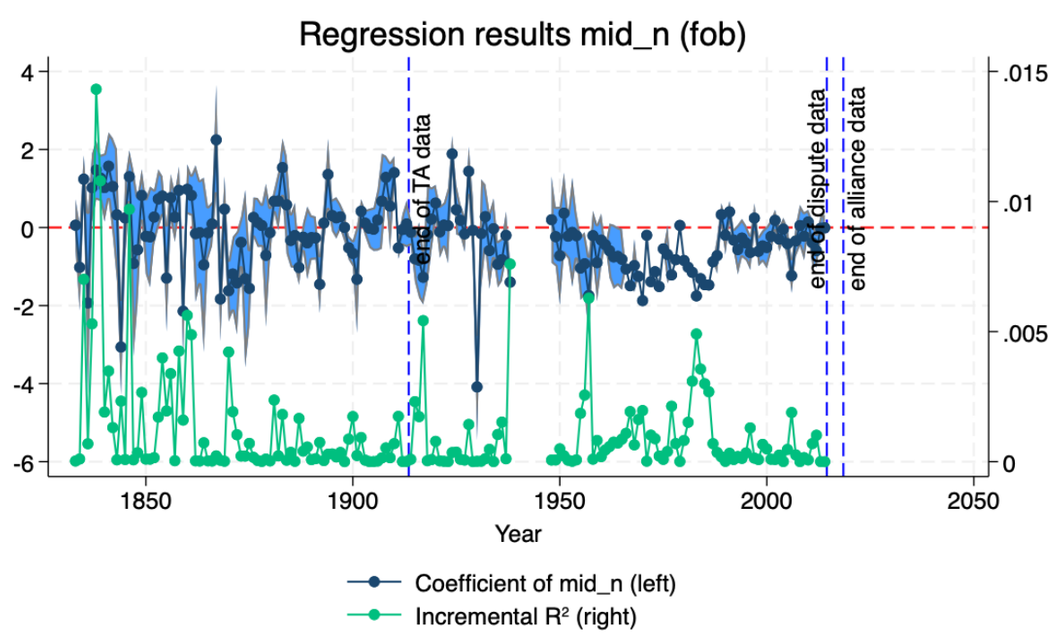

Regression results atop_allie (fob)
end of TA data
end of dispute data
end of alliance data
Year
Coefficient of atop_allie (left)
Incremental R² (right)

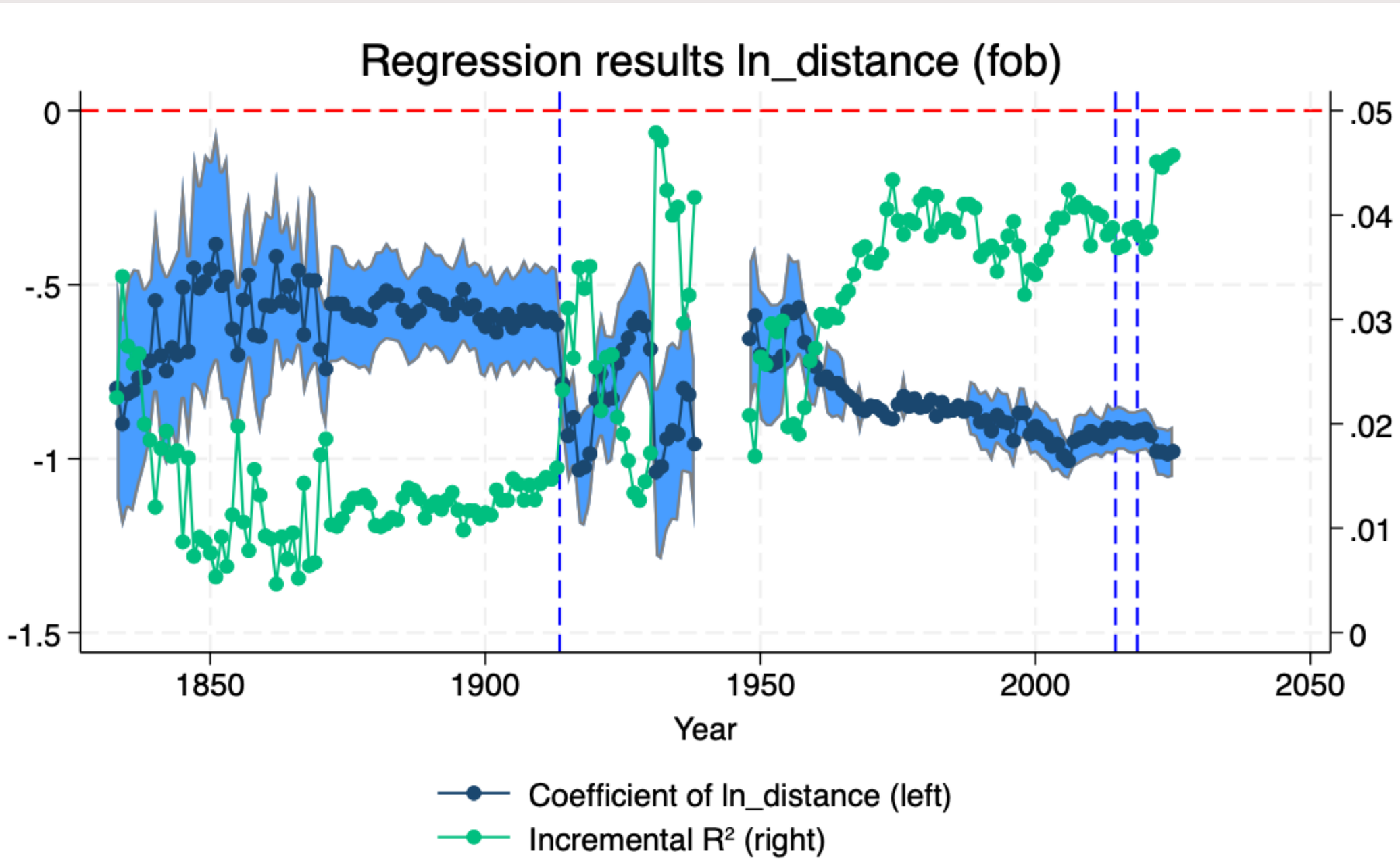

Regression results ln_distance (fob)
Year
Coefficient of ln_distance (left)
Incremental R² (right)

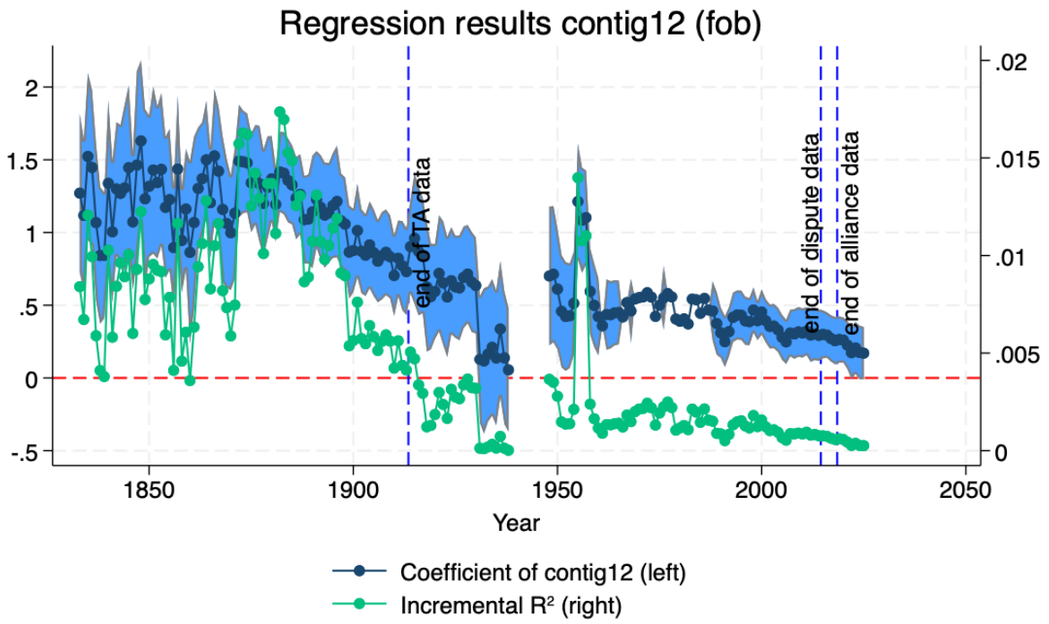

Regression results common_empire (fob)
end of TA data
end of dispute data
end of alliance data
Year
Coefficient of common_empire (left)
Incremental R² (right)


Regression results TA (fob)
2
1
0
-1
-2
.015
.01
.005
0
end of TA data
1840
1860
1880
1900
1920
Year
Coefficient of TA (left)
Incremental R² (right)


Regression results atop_allie (fob)
end of TA data
end of dispute data
end of alliance data
2
0
-2
-4
-6
-8
.006
.004
.002
0
1850
1900
1950
2000
2050
Year
Coefficient of atop_allie (left)
Incremental R² (right)

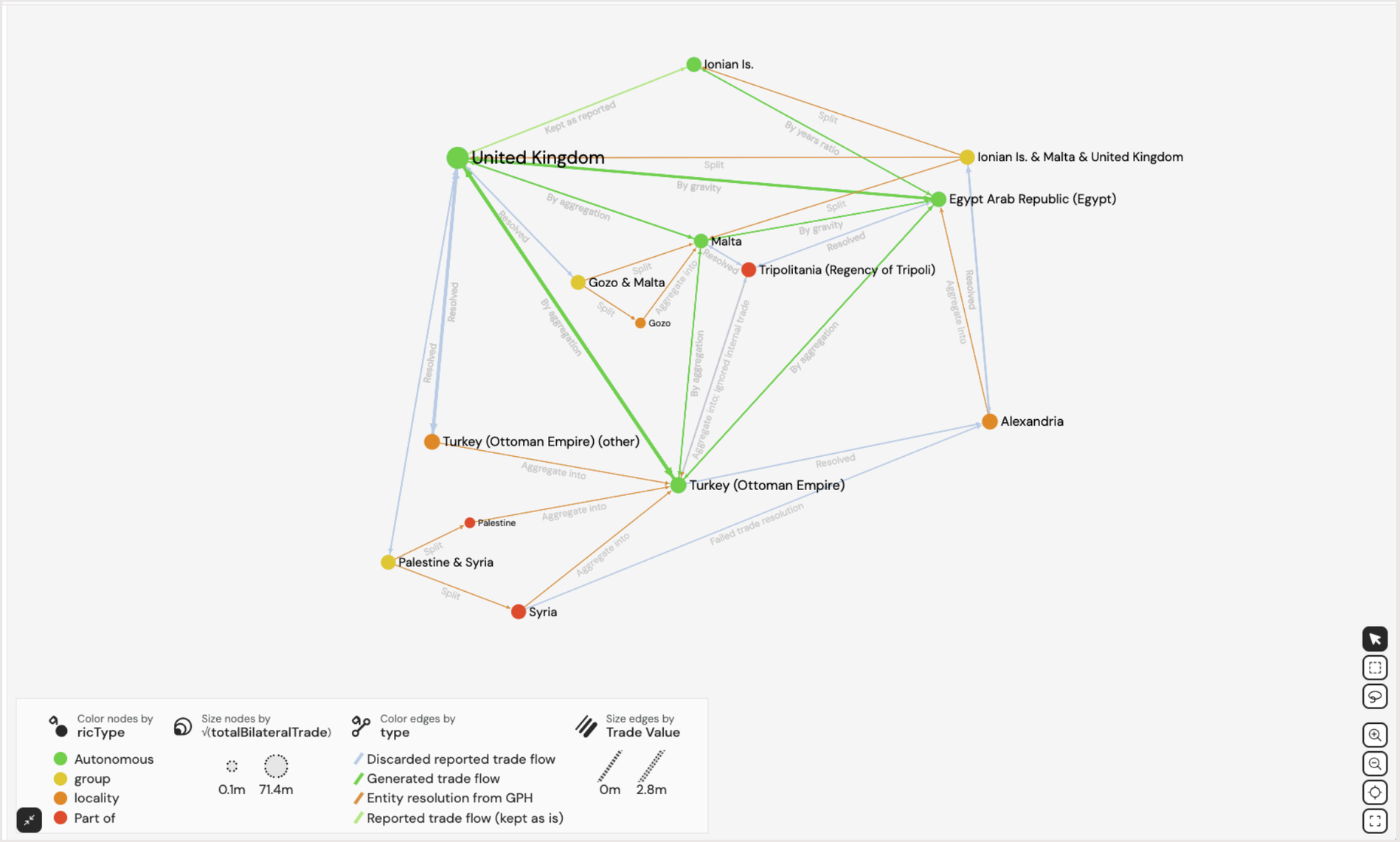

Regression results mid_n (fob)
end of TA data
end of dispute data
end of alliance data
4
2
0
-2
-4
-6
.015
.01
.005
0
1850
1900
1950
2000
2050
Year
Coefficient of mid_n (left)
Incremental R² (right)


Regression results ln_distance (fob)
0
-.5
-1
-1.5
.05
.04
.03
.02
.01
0
1850
1900
1950
2000
2050
Year
Coefficient of ln_distance (left)
Incremental R² (right)

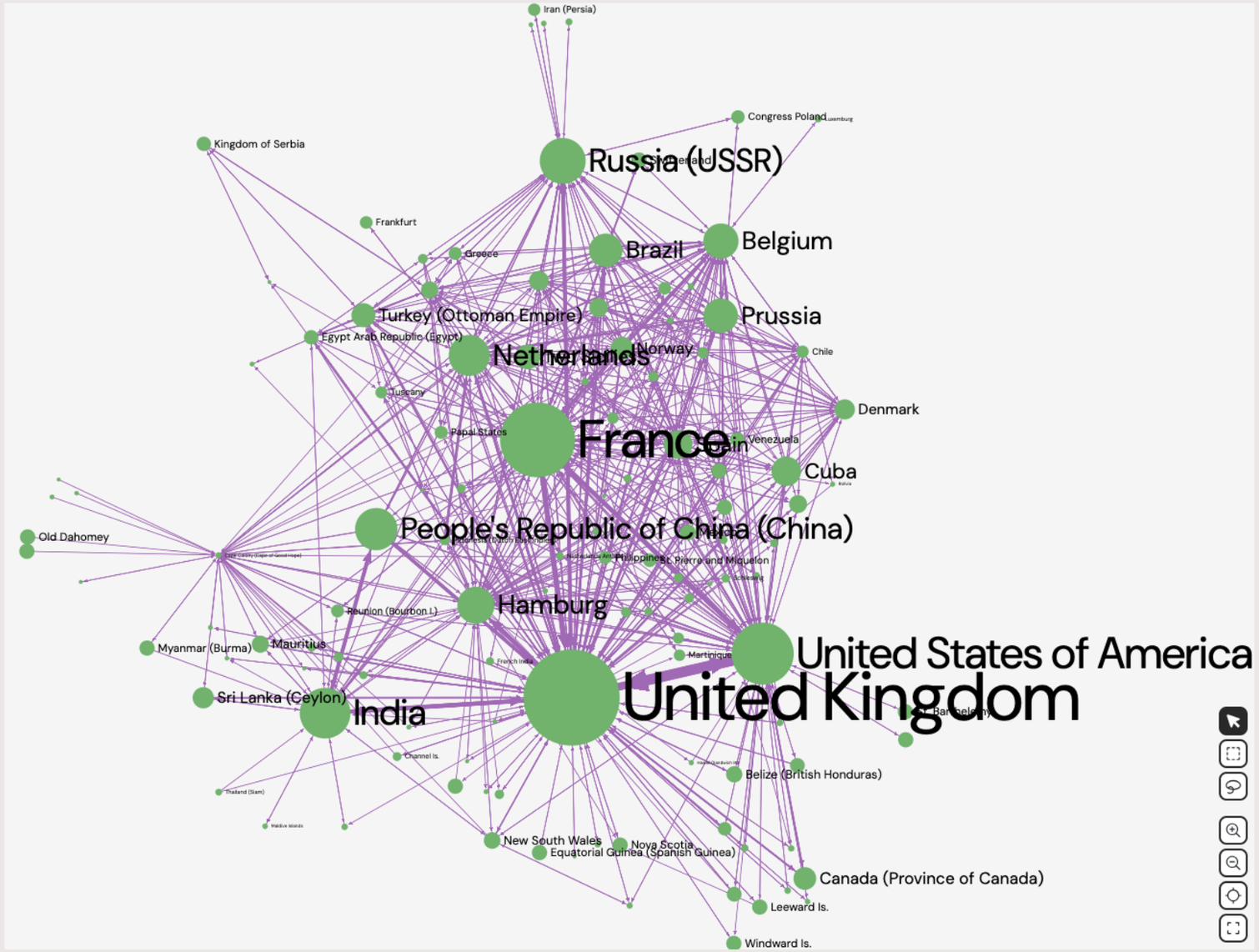

Regression results contig12 (fob)
2
1.5
1
.5
0
-.5
.02
.015
.01
.005
0
1850
1900
1950
2000
2050
Year
end of TA data
end of dispute data
end of alliance data
Coefficient of contig12 (left)
Incremental R² (right)


Regression results common_empire (fob)
4
3
2
1
0
.06
.04
.02
0
1850
1900
1950
2000
2050
Year
end of TA data
end of dispute data
end of alliance data
Coefficient of common_empire (left)
Incremental R² (right)

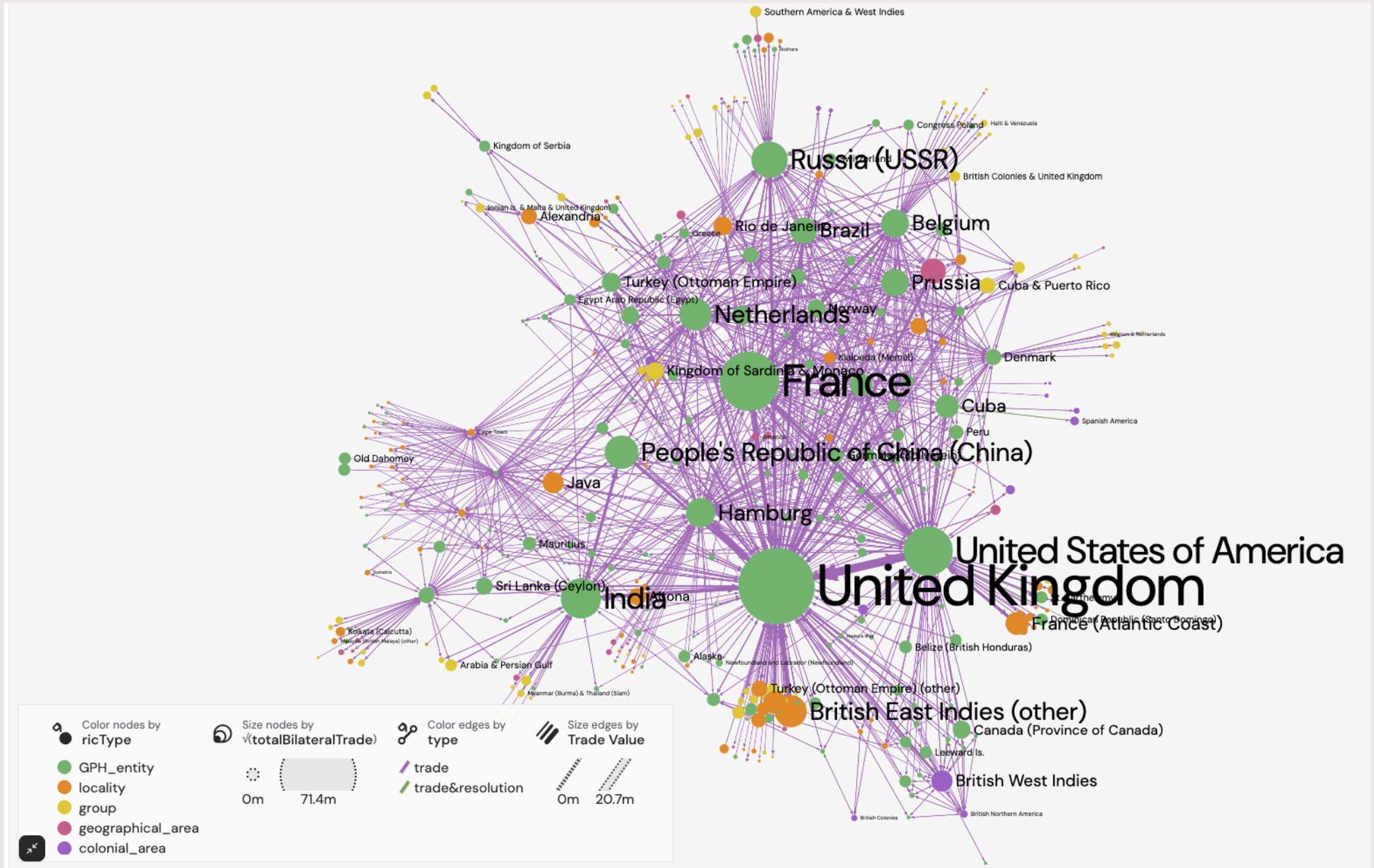
Examples of resolution around Malta, 1850



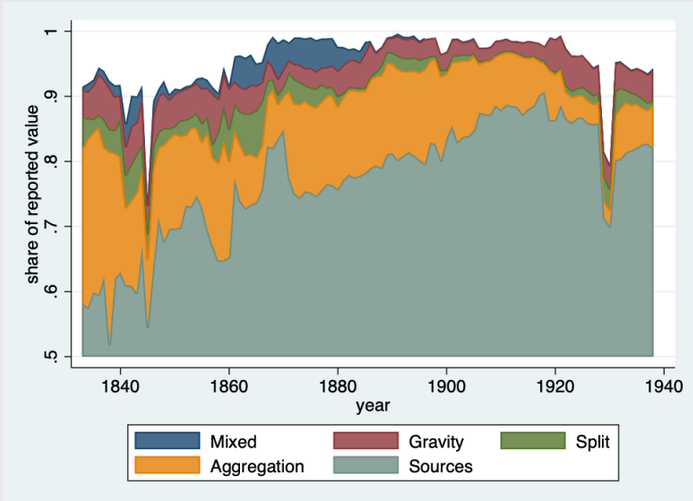
The 1850 trade network after the gravity step



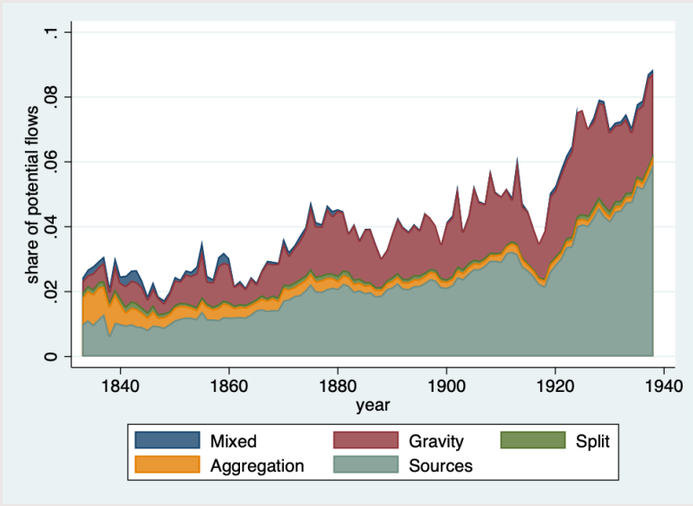
The 1850 trade network

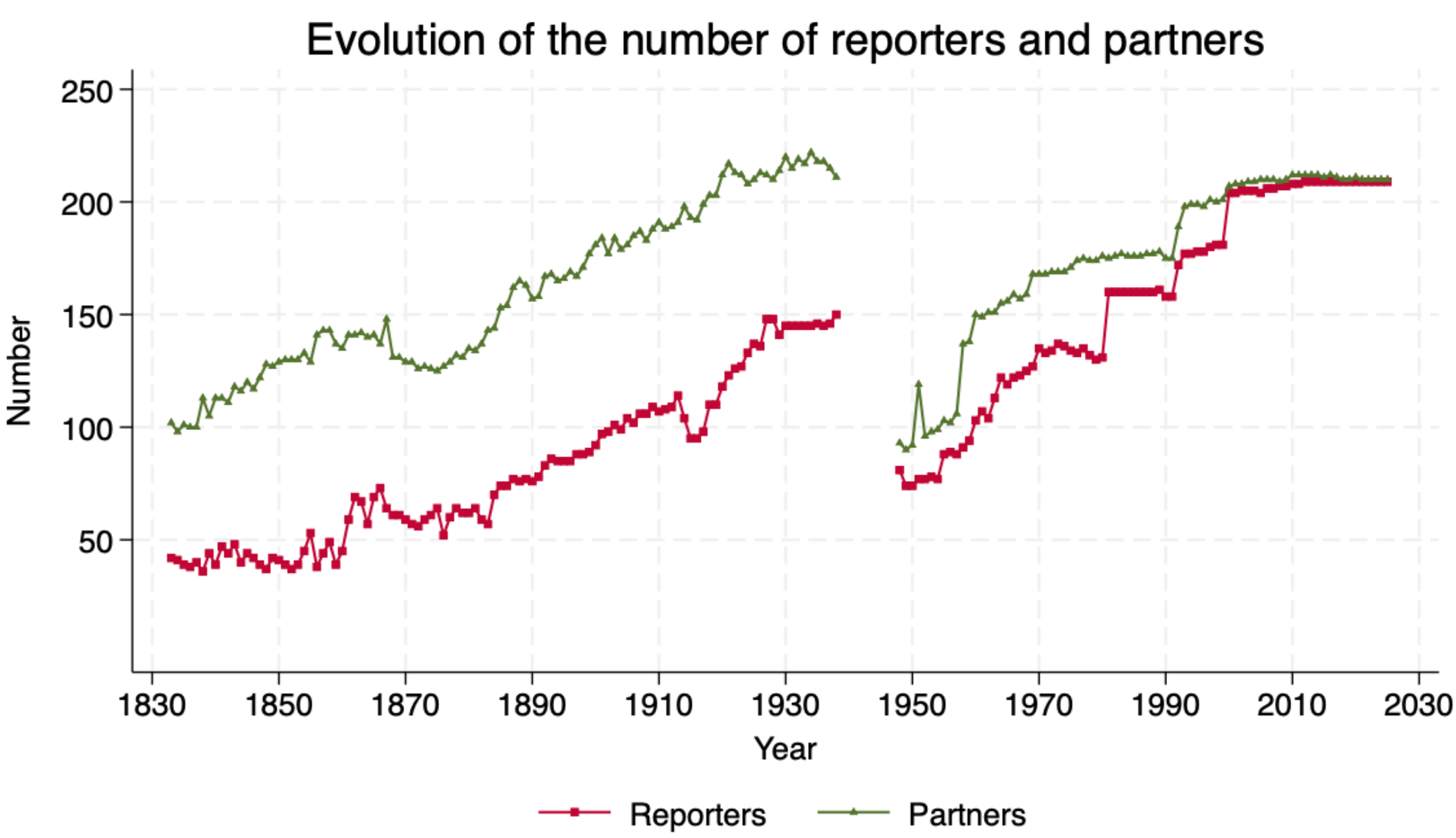


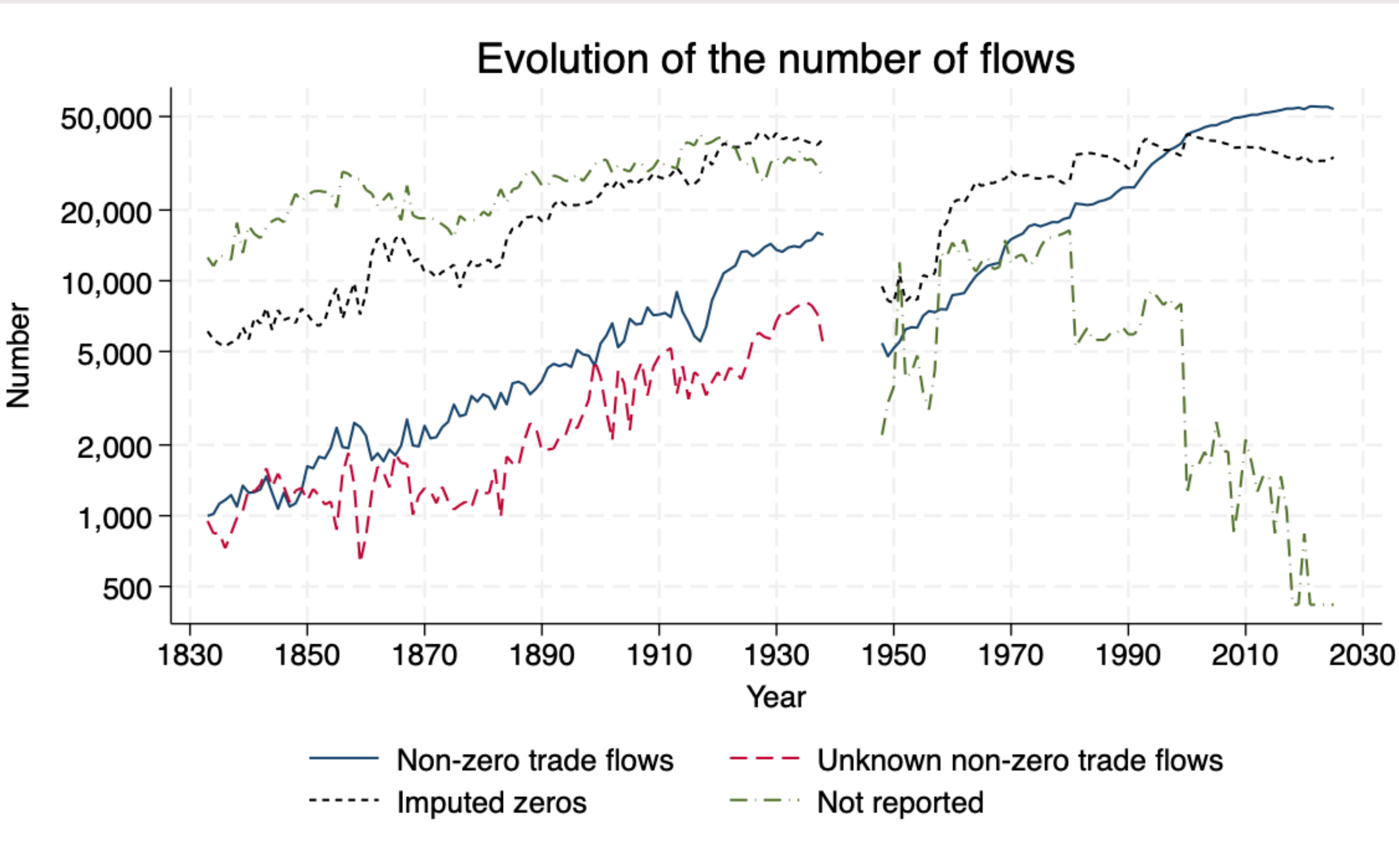
How much trade do we normalise? Value

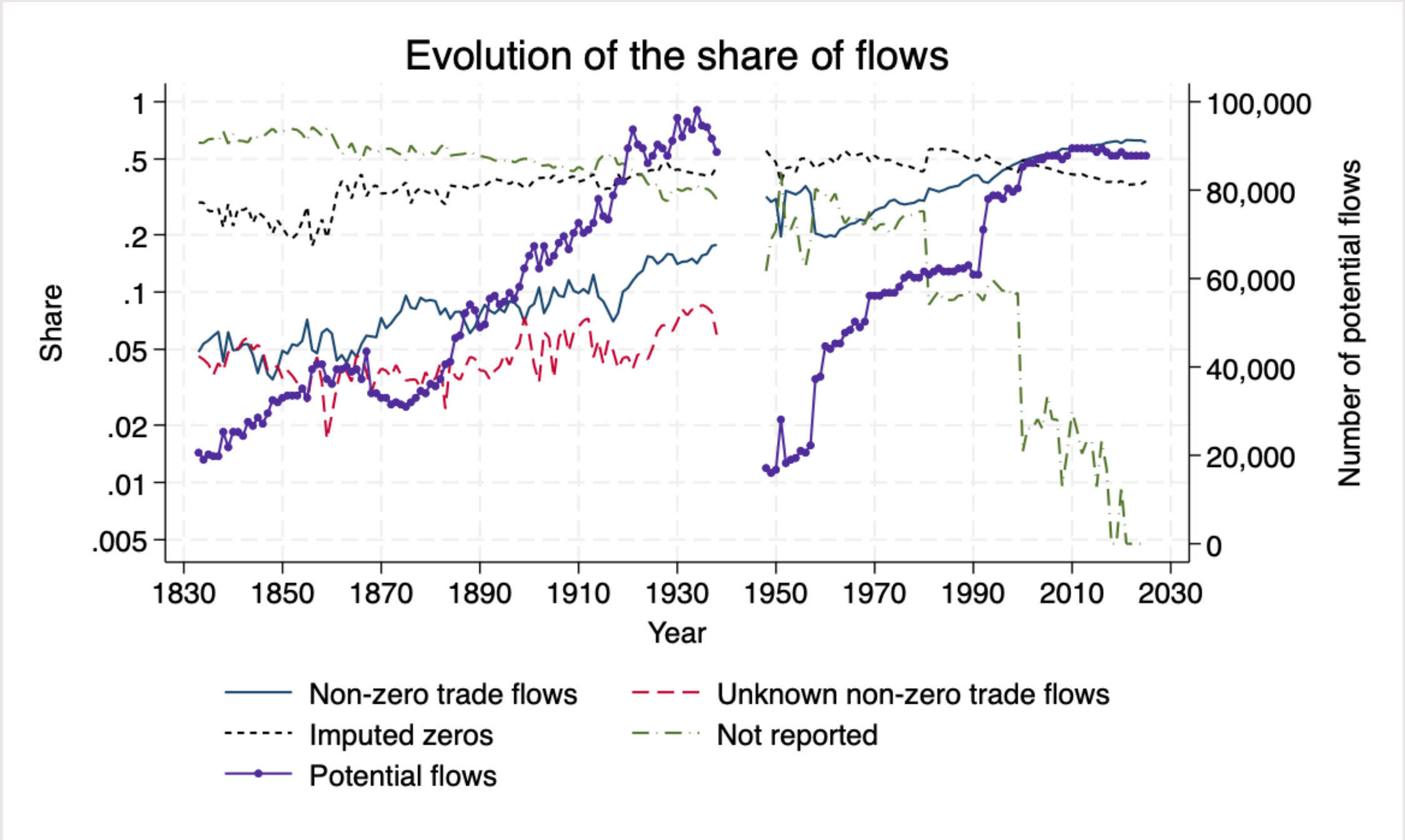


How much trade do we normalise? Flows




Evolution of the number of reporters and partners
Number
250
200
150
100
50
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2030
Year
Reporters
Partners


Evolution of the number of flows
Number
50,000
20,000
10,000
5,000
2,000
1,000
500
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2030
Year
Non-zero trade flows
Unknown non-zero trade flows
Imputed zeros
Not reported


Evolution of the share of flows
Share
1
.5
.2
.1
.05
.02
.01
.005
100,000
80,000
60,000
40,000
20,000
0
Number of potential flows
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2030
Year
Non-zero trade flows
Unknown non-zero trade flows
Imputed zeros
Not reported
Potential flows

II. IDENTIFYING TRADE COMMUNITIES

IDENTIFYING TRADE COMMUNITIES (1)

« The problem of community detection requires the partition of a network into communities of densely connected nodes, with the nodes belonging to different communities being only sparsely connected » Blondel et al. (2008)

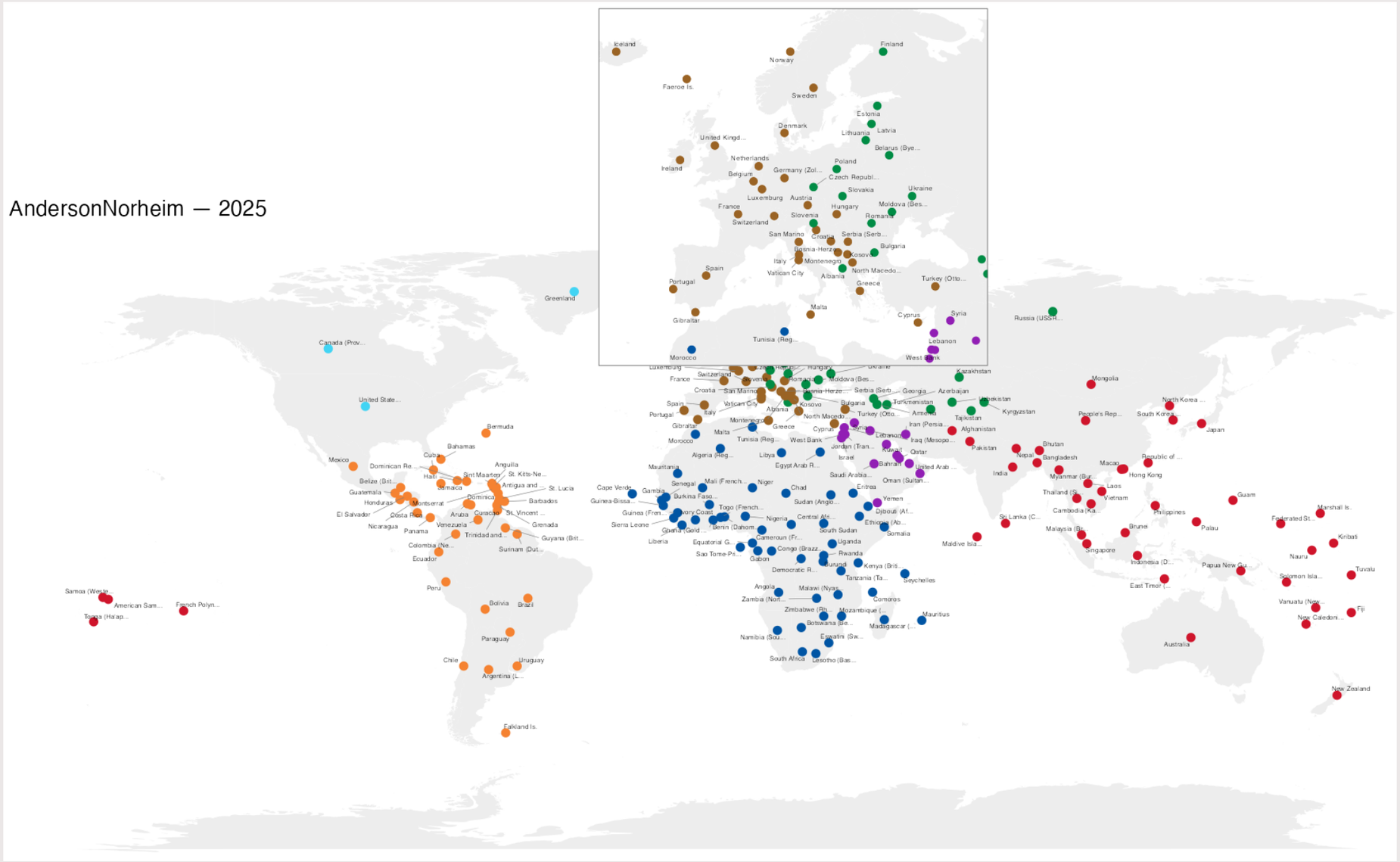
Trade studies: Either exogenous, sometimes endogenous, such as IntraMax

Poon (1997)

Network community has worked a lot on the subject

Blondel et al. (2008) : Louvain

Anderson and Norheim (1993)’s blocs as a reference



IDENTIFYING TRADE COMMUNITIES (3)

Two issues:

1. What is the algorithm?
2. What is the weight/proximity mesure?

IDENTIFYING TRADE COMMUNITIES (4) ALGORITHMS

Intramax algorithm:

1. Computes all proximities between actors/groups
2. Regroup the closest pair
3. Repeat till the regrouping increases internal trade in the group by more than 10% of world trade

Louvain algorithm:

Louvain adds a global measure grouping quality "modularity" (Newman (2006))

$$Q = \frac{1}{2m} . \sum_{i,j} \left(A_{i,j} . \frac{k_i . k_j}{2m} . \delta(c_i, c_j) \right)$$

$$Q \in [-0.5, 1]$$

Which is the ratio between the weight of links inside groups compared to random links

M. E. J. Newman, « Modularity and community structure in networks », *Proc. Natl. Acad. Sci. USA*, vol. 103, no 23, 2006, p. 8577–8582 https://dx.doi.org/10.1073%2Fpnas.0601602103

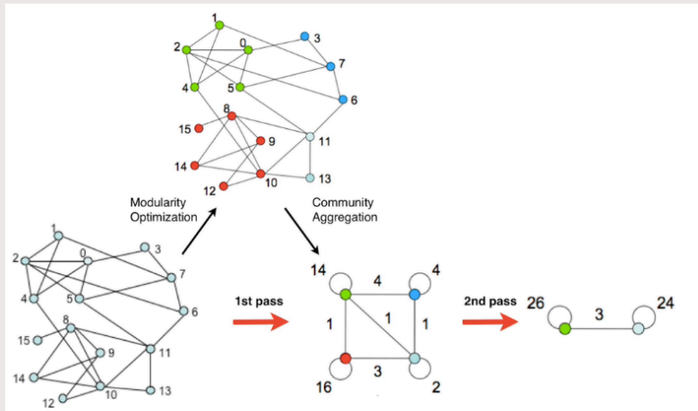


Figure 1. Visualization of the steps of our algorithm. Each pass is made of two phases: one where modularity is optimized by allowing only local changes of communities; one where the found communities are aggregated in order to build a new network of communities. The passes are repeated iteratively until no increase of modularity is possible.

IDENTIFYING TRADE COMMUNITIES (5) WEIGHT/PROXIMITY

We can have up to four flows by pair.

But we cannot simply look at exports or imports because of the large number of missing observations

In all cases, we compute 1 to 4 proximities and we take the max

$$Max\left(P_{ij}^{X};P_{ji}^{X};P_{ij}^{M};P_{ji}^{M}\right)=A_{ij}=A_{ji}$$

1. We can just take raw trade

2. Intramax proximity

One looks for the polity/bloc pair that is the furthest from a random model of bilateral trade.
X_{ij} is (eg) fob exports from polity/bloc i to polity/bloc j
X'_{ij} is counterfactual, random, no friction trade, defined as:

$$X'_{ij}=\frac{X_{i.}}{X}\frac{X_{.j}}{X}X=\frac{X_{i.}X_{.j}}{X}$$

Weight/Proximity: $(X_{ij}-X'_{ij})+(X_{ji}-X'_{ji})[+(M_{ij}-M'_{ij})+(M_{ji}-M'_{ji})]$

- Poon's maximisation objective is in monetary terms, and as such gives an advantage to large polity pairs.
  - We believe if a small polity does 100% of its trade with a big one, even if this trade is small in monetary terms, they must be in the same bloc. So we normalise. Kohl & Brouwer (2014)

$$\text{Proximity} = \frac{X_{ij} - X'_{ij}}{X_{i.}}, \frac{X_{ji} - X'_{ji}}{X_{j.}}, \frac{M_{ij} - M'_{ij}}{M_{.j}} and \frac{M_{ji} - M'_{ji}}{M_{.I}}$$

3. TIBI (Trade Intensity Bilateral Index)

Poon's maximisation objective can also be expressed as a ratio : CEPII's Relative bilateral intensity of port Intensity:

$$X_{ij}/X'_{ij} = (X_{ij}/X_{i.})/(X_{.j}/X)) \in [-1, X/X_{.j}].$$

Inspired by Balassa (1965)'s index of comparative advantage. Eg Gaulier and Ünal-Kesenci (2004)

Iapadre (2006), Daudin et al. (2011) mention three issues:

Solving range variability

Solution: $I^1{}_{ij} = (X_{ij}/X_{i.})/(X_{rj}/X)) \in [0, +\infty[$

Solving range asymmetry

Solution: $I^2{}_{ij} = (I^1{}_{ij} - 1)/(I^1{}_{ij} + 1) \in [-1, 1]$

Solving dynamic ambiguity

Solution: $TIBI_{ij} = (I^1{}_{ij}/I^3{}_{ij} - 1)/(I^1{}_{ij}/I^3{}_{ij} + 1) \in [-1, 1]$

Where: $I^3 = (1 - (X_{ij}/X_{i.}))/(1 - (X_{rj}/X))$

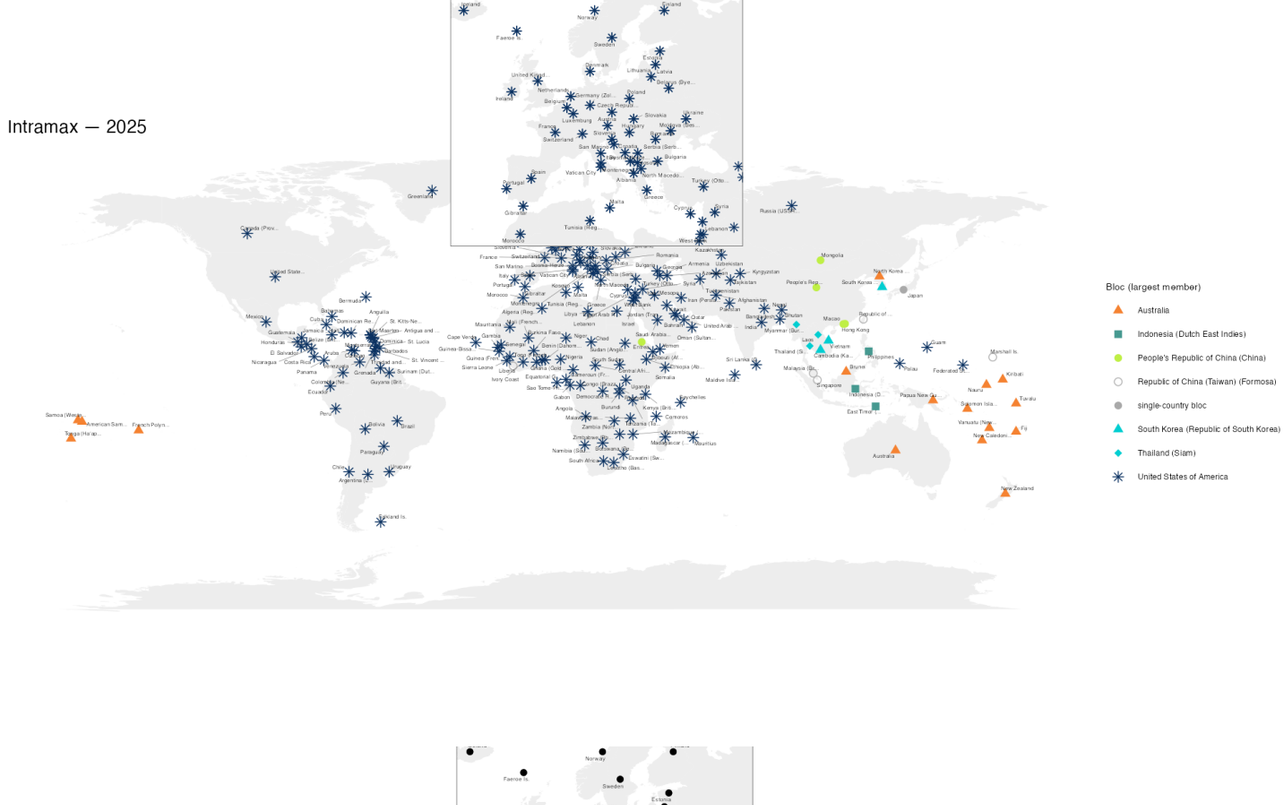
Proximity measures in 1938

expCountry	impCountry	Poon1	Poon2	REI	I1	I2	I3	TIBI
United Kingdom	India	27,102,807	0.24	5.07	19.25	0.90	0.71	0.93
Canada (Province of Canada)	United States of America	54,593,186	0.31	2.73	6.05	0.72	0.56	0.83
United States of America	Canada (Province of Canada)	49,856,880	0.19	2.16	6.33	0.73	0.69	0.80
France	Belgium	18,704,330	0.29	4.18	6.05	0.72	0.66	0.80
Netherlands	Belgium	7,244,770	0.14	2.58	2.93	0.49	0.83	0.56
India	United Kingdom	17,237,537	0.27	1.71	1.83	0.29	0.55	0.54
United Kingdom	Netherlands	5,970,202	0.05	1.83	2.16	0.37	0.93	0.40
France	Netherlands	3,799,255	0.06	1.92	2.11	0.36	0.93	0.39
United Kingdom	France	6,673,710	0.06	1.79	2.09	0.35	0.93	0.39
Netherlands	France	2,988,465	0.06	1.79	1.90	0.31	0.93	0.34
Netherlands	United Kingdom	7,292,692	0.14	1.38	1.42	0.17	0.76	0.31
United States of America	France	7,103,810	0.03	1.36	1.70	0.26	0.96	0.28
Canada (Province of Canada)	United Kingdom	16,696,877	0.09	1.25	1.36	0.15	0.81	0.25
United Kingdom	Canada (Province of Canada)	4,344,973	0.04	1.24	1.29	0.13	0.95	0.16
United States of America	Netherlands	2,763,814	0.01	1.16	1.28	0.12	0.98	0.13
United States of America	United Kingdom	4,983,666	0.02	1.05	1.08	0.04	0.95	0.06
United Kingdom	United States of America	392,914	0.00	1.02	1.02	0.01	1.00	0.01
India	France	-201,969	0.00	0.96	0.95	-0.02	1.00	-0.03
India	United States of America	-1,106,690	-0.02	0.90	0.90	-0.06	1.02	-0.07
France	United States of America	-1,655,569	-0.03	0.86	0.85	-0.08	1.03	-0.10
India	Netherlands	-797,324	-0.01	0.81	0.79	-0.12	1.01	-0.12
United Kingdom	Belgium	-1,986,071	-0.02	0.81	0.78	-0.12	1.02	-0.14
France	United Kingdom	-3,712,237	-0.06	0.85	0.84	-0.09	1.10	-0.14
India	Belgium	-2,512,095	-0.04	0.57	0.55	-0.29	1.05	-0.31
United States of America	Belgium	-8,424,108	-0.03	0.65	0.54	-0.30	1.06	-0.32
Netherlands	United States of America	-4,822,450	-0.10	0.46	0.45	-0.38	1.13	-0.43
United States of America	India	-8,882,670	-0.03	0.43	0.33	-0.50	1.06	-0.52
Netherlands	India	-2,077,787	-0.04	0.30	0.29	-0.55	1.05	-0.57
France	India	-2,776,845	-0.04	0.28	0.26	-0.59	1.05	-0.61
Canada (Province of Canada)	Netherlands	-8,525,517	-0.05	0.24	0.19	-0.68	1.07	-0.69
Canada (Province of Canada)	Belgium	-13,026,826	-0.07	0.19	0.15	-0.74	1.11	-0.77
India	Canada (Province of Canada)	-8,806,570	-0.14	0.15	0.14	-0.75	1.18	-0.78
Canada (Province of Canada)	France	-11,695,530	-0.07	0.12	0.09	-0.83	1.10	-0.85
France	Canada (Province of Canada)	-9,490,448	-0.15	0.09	0.09	-0.84	1.19	-0.87
Netherlands	Canada (Province of Canada)	-7,415,260	-0.15	0.09	0.08	-0.84	1.19	-0.87
Canada (Province of Canada)	India	-9,552,616	-0.05	0.08	0.07	-0.88	1.08	-0.88

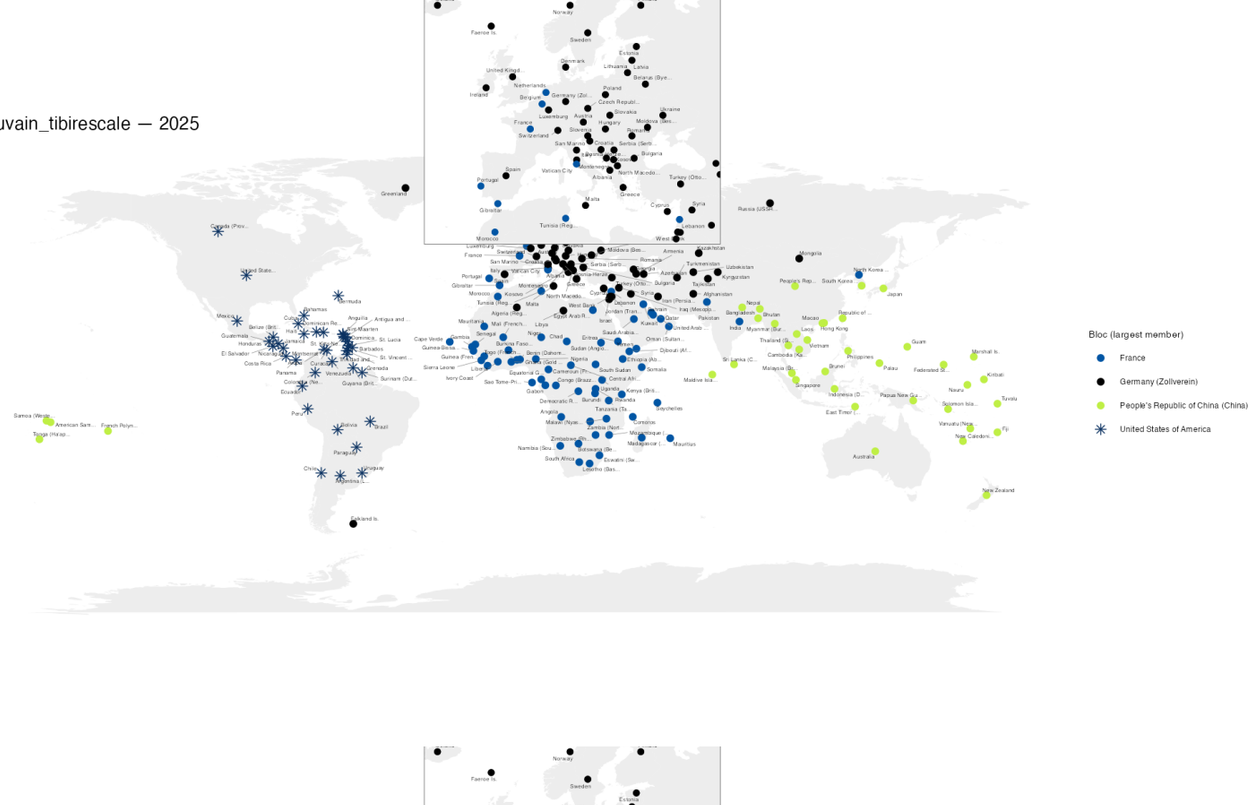
Identifying trade communities (6)

1. Intramax with Poon 2 measure ($\frac{X_{ij}-X'_{ij}}{X_{i.}}$)
2. Louvain (only accepts positive weights)
   a. With raw trade
   b. With TIBI + 1
   Issue : missing and zeros are treated the same way
3. Signed Louvain with raw TIBI

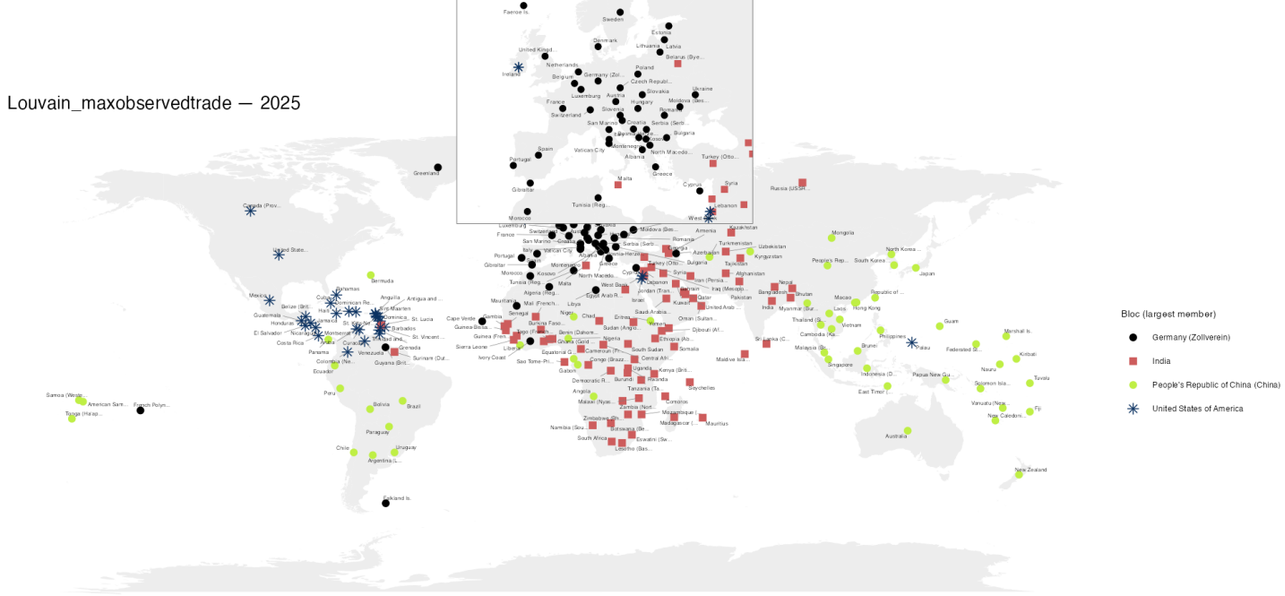
Intramax — 2025



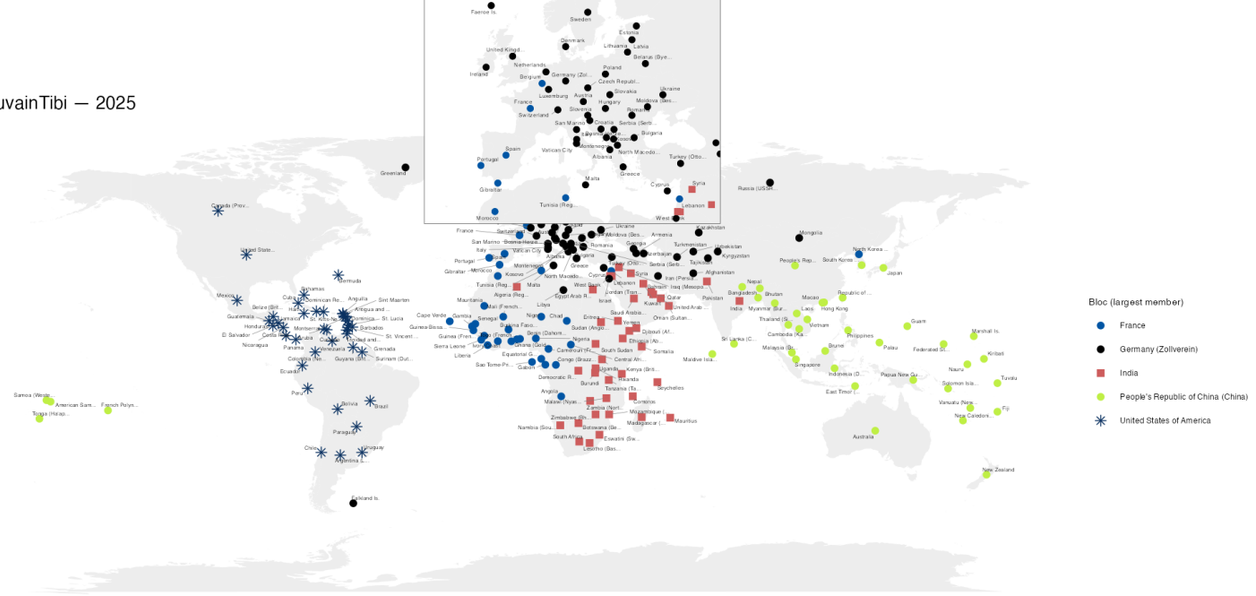
Louvain_tibirescale — 2025

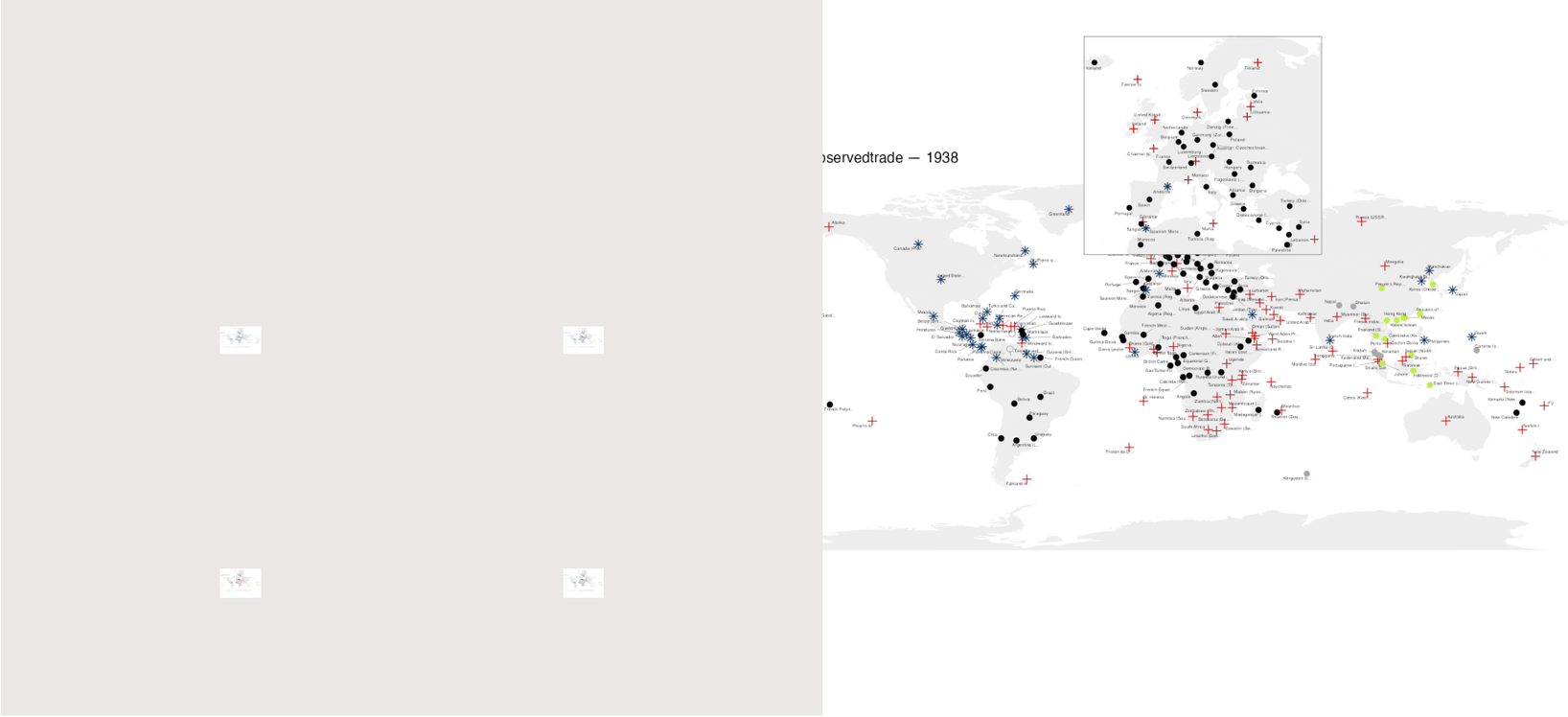


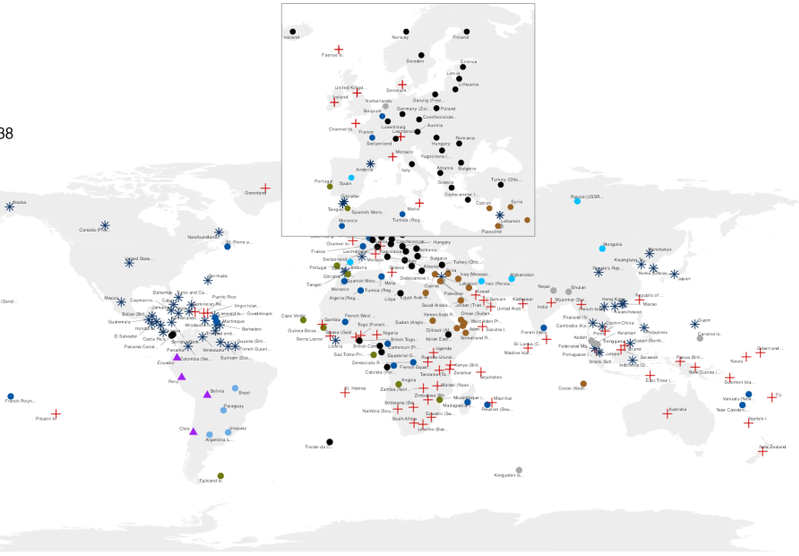
Louvain_maxobservedtrade — 2025

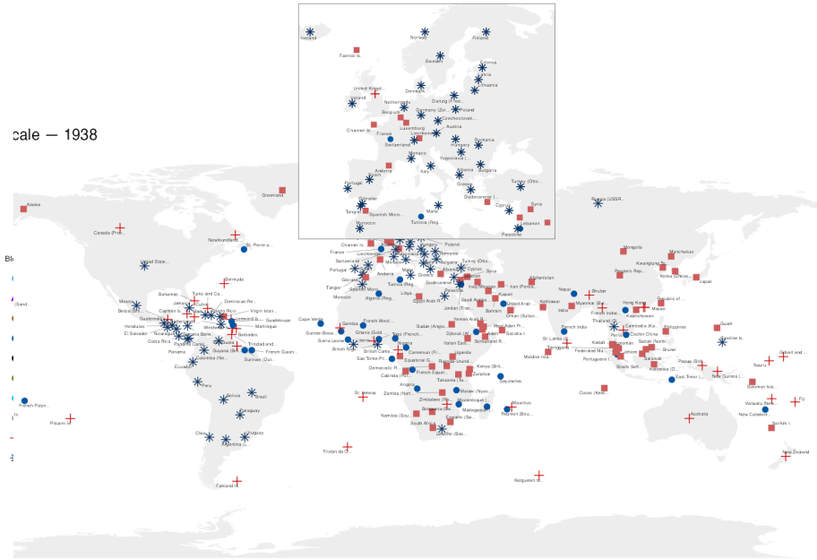


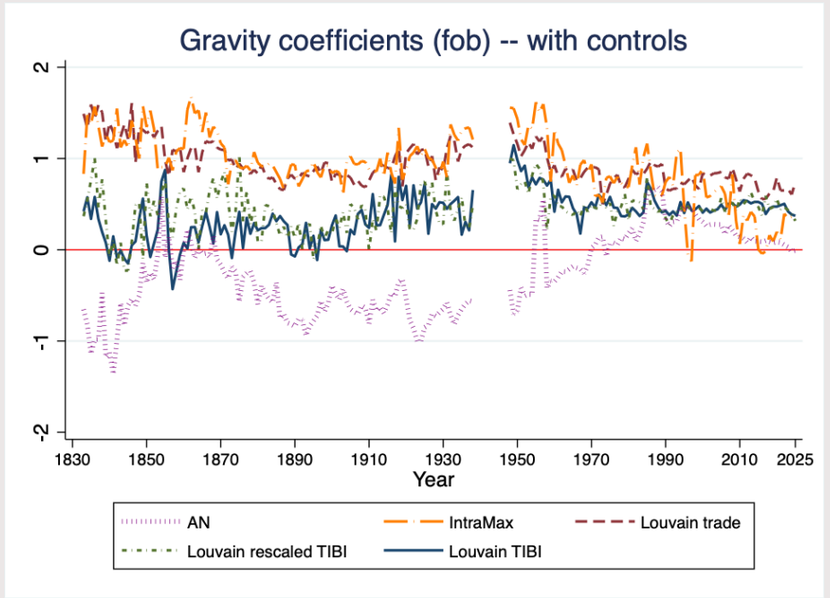
LouvainTibi — 2025

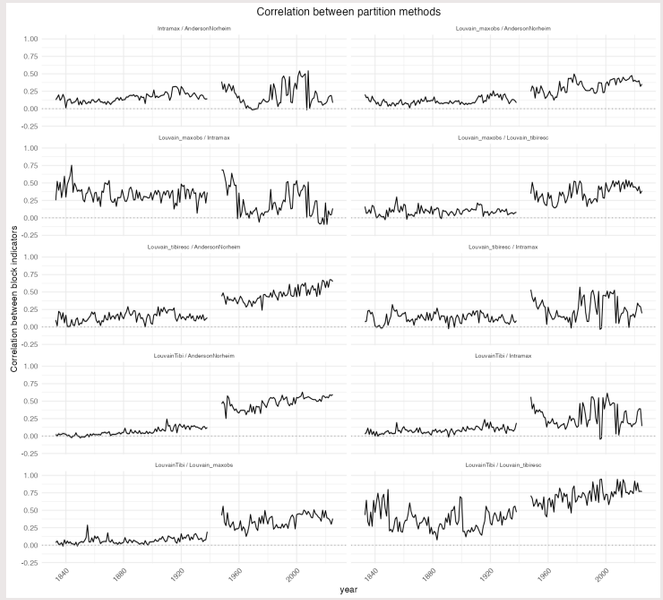


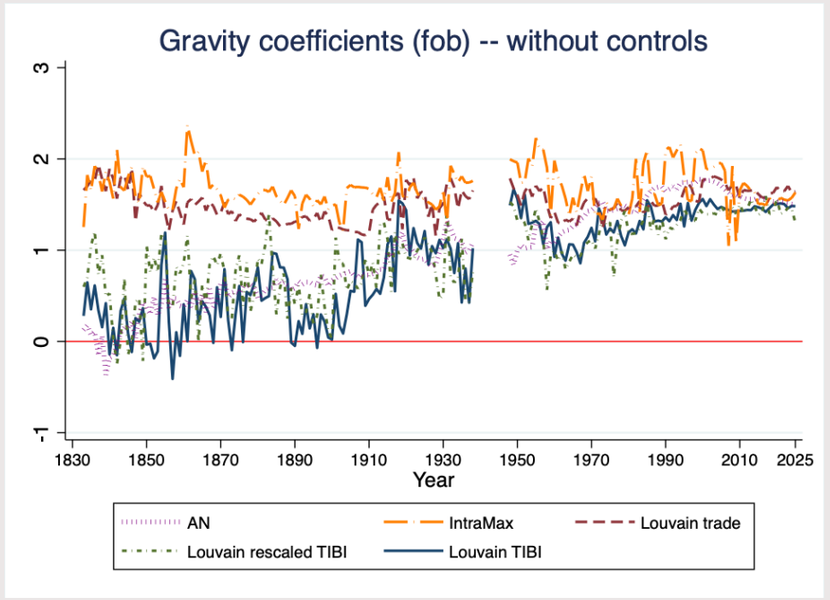

oservedtrade — 1938
Bloc (Iar
Iceland
Norway
Finland
Faeroe Is.
Sweden
Estonia
Latvia
Lithuania
United Kingd...
Ireland
Denmark
Netherlands
Danzig (Free...
Belgium
Germany (Zol...
Poland
Czechoslovak...
Luxemburg
Austria
Channel Is.
France
Liechtenst...
Switzerland
Hungary
Romania
Yugoslavia (...
Monaco
Albania
Bulgaria
Andorra
Italy
Greece
Spain
Dodecanese I...
Turkey (Otto...
Portugal
Gibraltar
Cyprus
Syria
Tangier
Spanish Moro...
Malta
Tunisia (Reg...
Lebanon
Morocco
Palestine
Greenland
St. Pierre a...
Algeria (Reg...
Libya
Egypt Arab R...
Jordan (Tran...
Iraq (Mesopo...
Afghanistan
Iran (Persia...
Kuwait
Bahrain
Saudi Arabia...
United Arab ...
Yemen Arab R...
Oman (Sultan...
Aden
Somaliland R...
Italian East...
Nepal
Bhutan
Kathiawar
India
Myanmar (Bur...
Mongolia
Kwangtung Pr...
People's Reg...
Hong Kong
Kwanchowan
Thailand (Si...
Cambodia (Ka...
Cochin China
Kelantan
Perlis
Kedah
Trengganu
Sabah (Nor...
Brunei
Federated Ma...
Sarawak
Johore
Straits Sett...
Indonesia
Portuguese I...
Maldive Isla...
Sri Lanka (C...
French India
French Indoc...
Cape Verde
Gambia
French West ...
Togo (French...
Sudan (Anglo...
Djibouti (Af...
West Aden Pr...
Socotra I.
Guinea-Bissa...
Ghana (Gold ...
Nigeria
Sierra Leone
Liberia
Cameroun (Fr...
British Togo...
British Came...
Equatorial G...
Italian East...
Somaliland R...
Uganda
Sao Tome-Pri...
Democratic R...
Ruanda-Urund...
Kenya (Brit...
Zanzibar
Cabinda (Por...
French Equat...
Tanzania (Ta...
Seychelles
Angola
Malawi (Nyas...
Zambia (Nor...
Mozambique (...
Madagascar (...
Mauritius
Reunion (Bou...
Zimbabwe (Rh...
Namibia (Sou...
Botswana (Be...
South Africa
Eswatini (Sw...
Lesotho (Bas...
Cocos (Keel...
Tristan da C...
Kerguelen Is...
Puerto Rico
Virgin Islan...
Leeward Is.
Guadeloupe
Martinique
Barbados
Trinidad and...
Windward Is.
French Guian...
Suriname (Dut...
Brazil
Paraguay
Uruguay
St. Helena

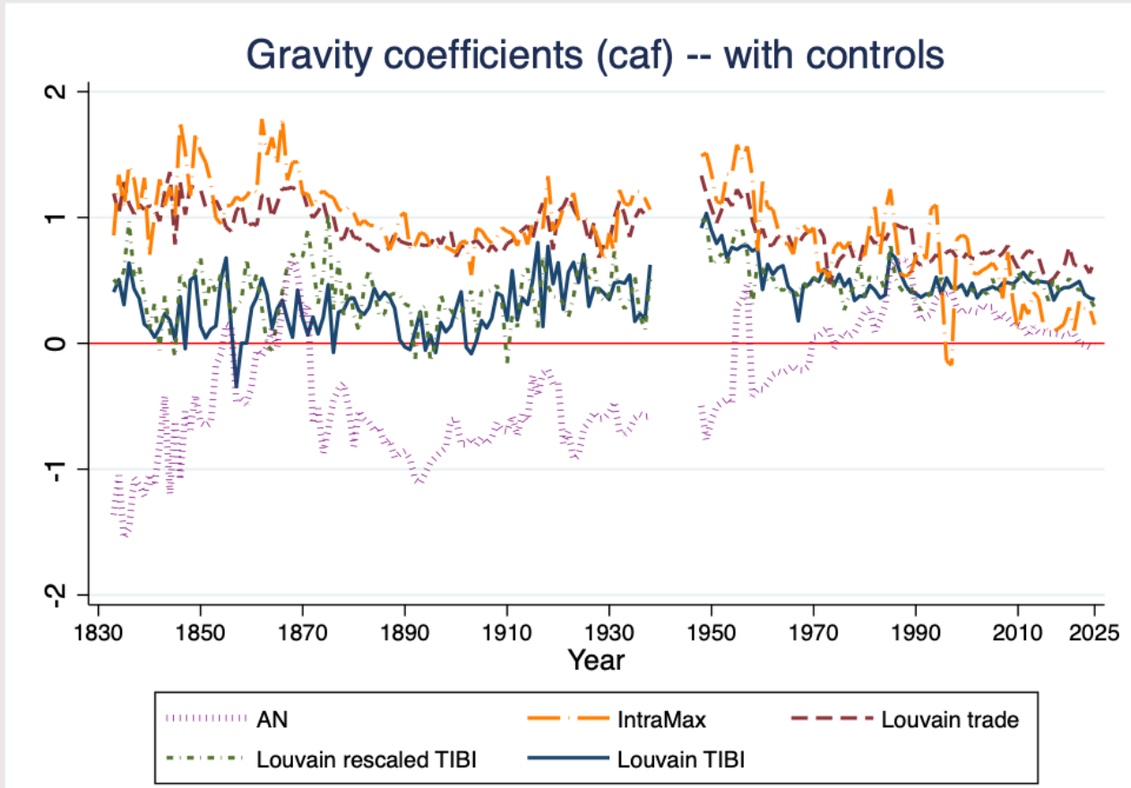

38

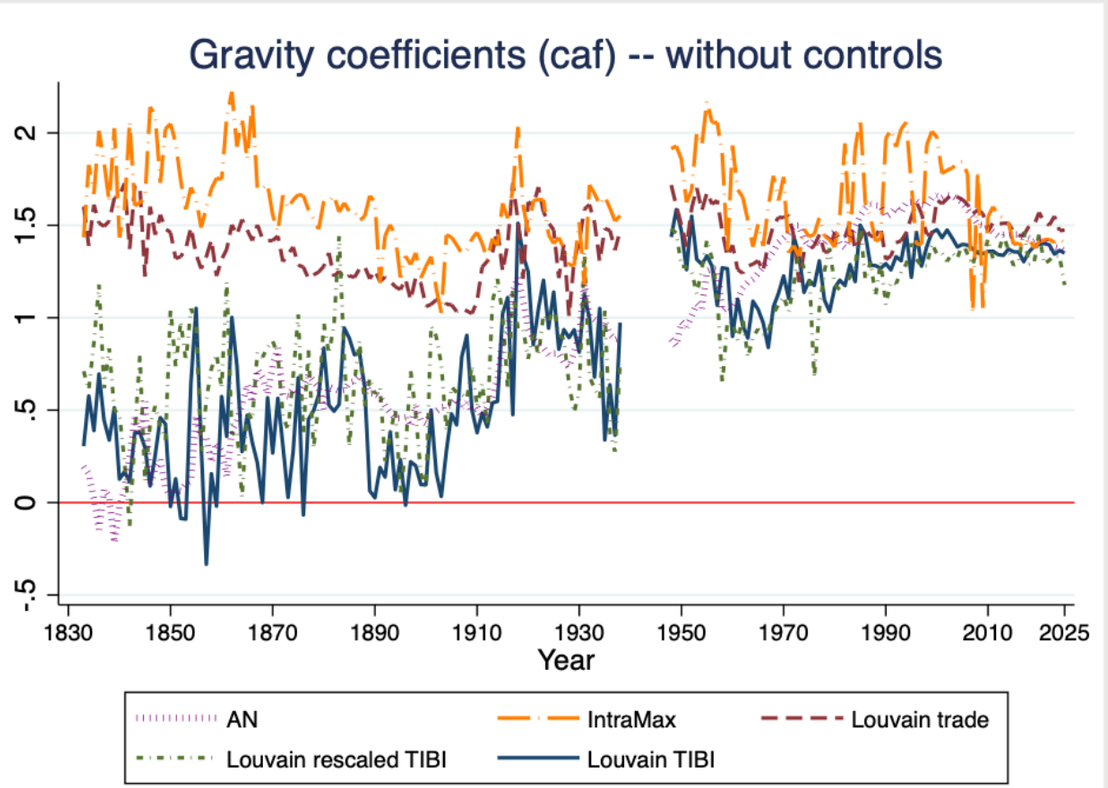

cale — 1938

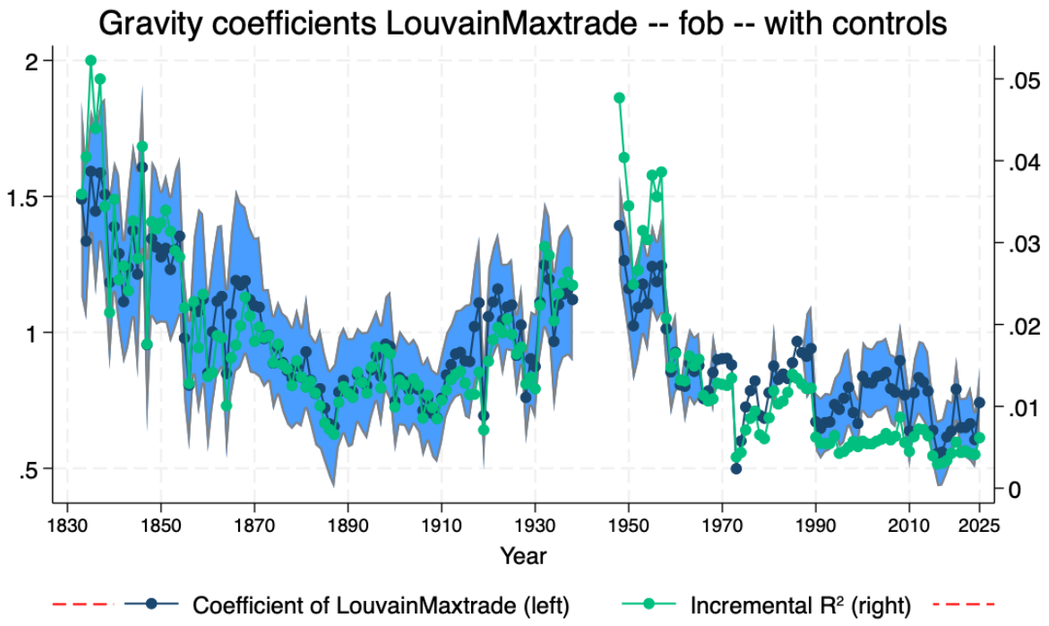

Gravity coefficients (fob) -- with controls
2
1
0
-1
-2
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
AN
IntraMax
Louvain trade
Louvain rescaled TIBI
Louvain TIBI

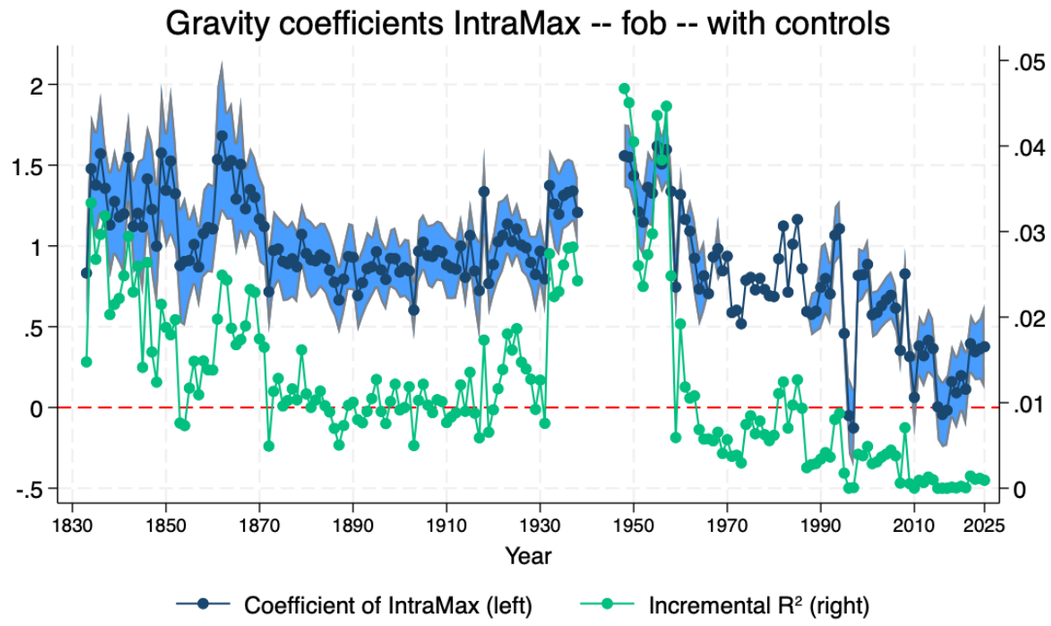

Correlation between partition methods
Intramax / AndersonNorheim
Louvain_maxobs / AndersonNorheim
Louvain_maxobs / Intramax
Louvain_maxobs / Louvain_tibiresc
Louvain_tibiresc / AndersonNorheim
Louvain_tibiresc / Intramax
LouvainTibi / AndersonNorheim
LouvainTibi / Intramax
LouvainTibi / Louvain_maxobs
LouvainTibi / Louvain_tibiresc
Correlation between block indicators
year


Gravity coefficients (fob) -- without controls
3
2
1
0
-1
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
AN
IntraMax
Louvain trade
Louvain rescaled TIBI
Louvain TIBI


Gravity coefficients (caf) -- with controls
2
1
0
-1
-2
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
AN
IntraMax
Louvain trade
Louvain rescaled TIBI
Louvain TIBI


Gravity coefficients (caf) -- without controls
2
1.5
1
.5
0
-.5
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
AN
IntraMax
Louvain trade
Louvain rescaled TIBI
Louvain TIBI

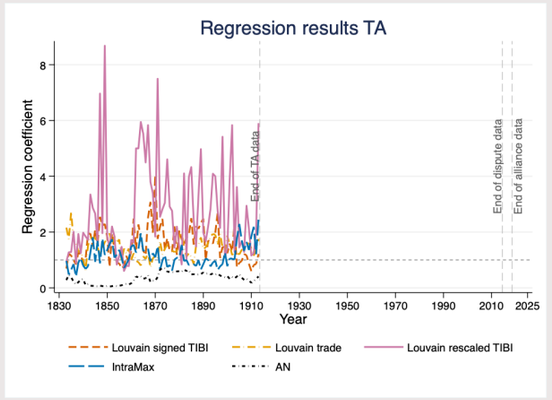

Gravity coefficients LouvainMaxtrade -- fob -- with controls
2
1.5
1
.5
.05
.04
.03
.02
.01
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
Coefficient of LouvainMaxtrade (left)
Incremental R² (right)

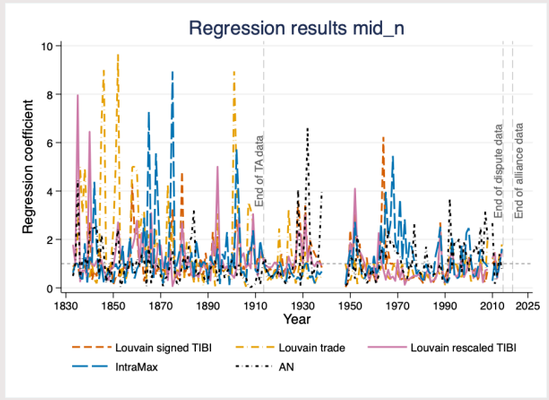

Gravity coefficients IntraMax -- fob -- with controls
2
1.5
1
.5
0
-.5
.05
.04
.03
.02
.01
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
Coefficient of IntraMax (left)
Incremental R² (right)

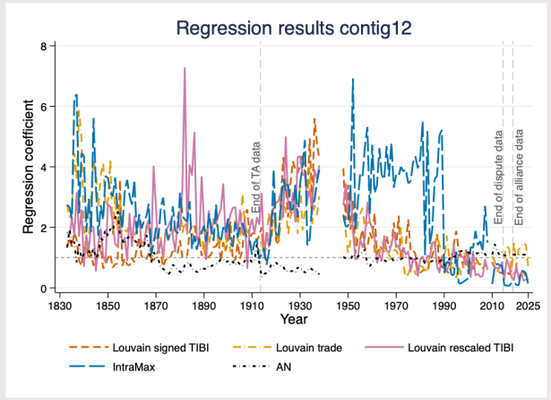
IV. DETERMINANTS OF TRADE COMMUNITIES

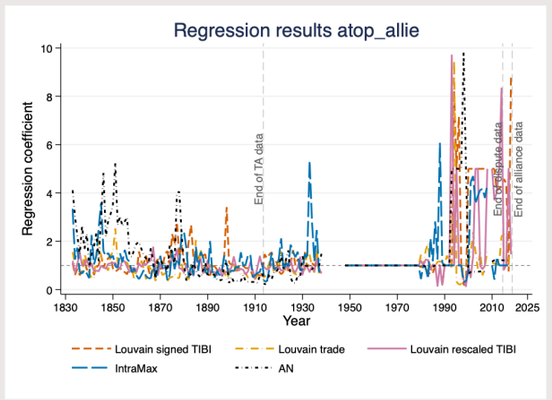
Logit regression on polity-pairs each year, with the usual control variables

No causality, obviously\

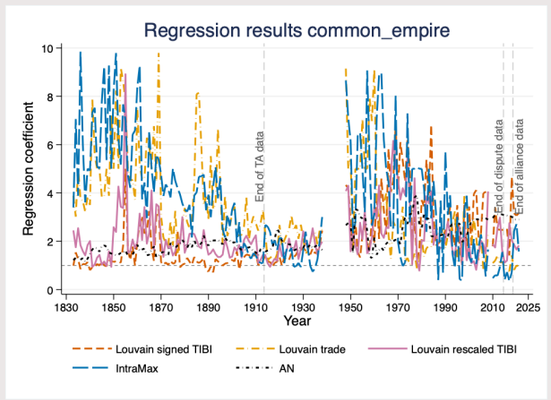

Regression results TA
Regression coefficient
0
2
4
6
8
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

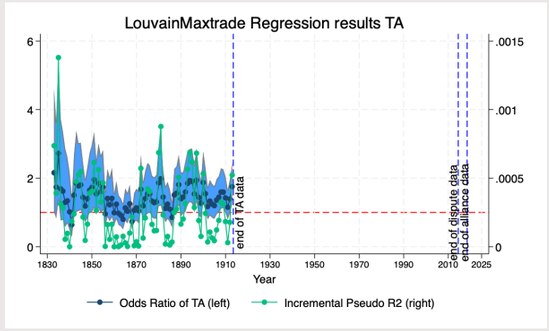

Regression results mid_n
Regression coefficient
0
2
4
6
8
10
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

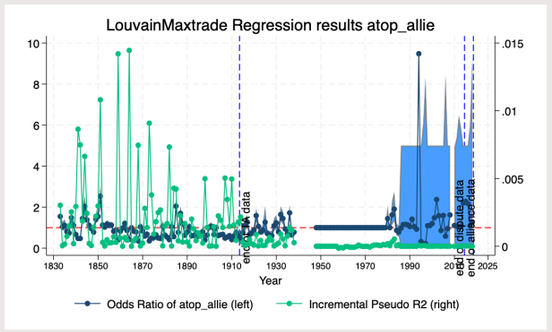

Regression results contig12
Regression coefficient
0
2
4
6
8
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

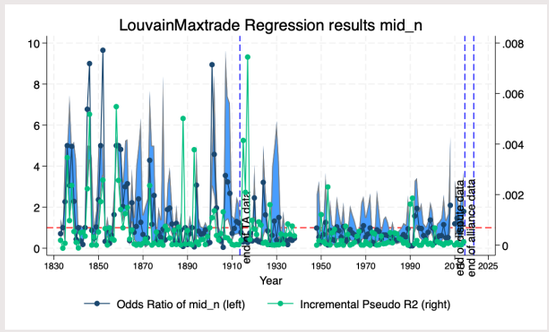

Regression results atop_allie
Regression coefficient
0
2
4
6
8
10
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

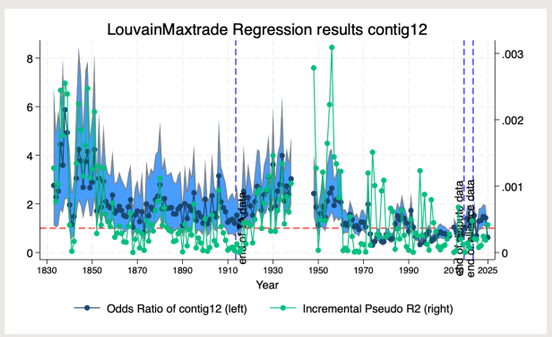

Regression results ln_distance
Regression coefficient
0
.5
1
1.5
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

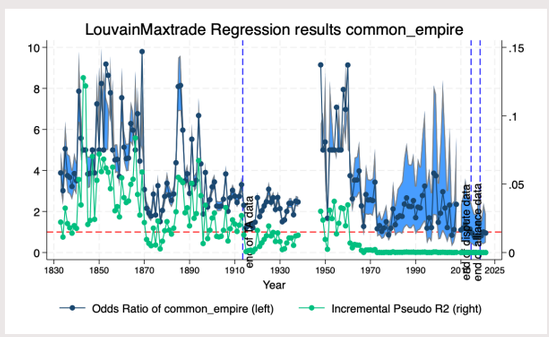

Regression results common_empire
Regression coefficient
0
2
4
6
8
10
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
End of TA data
End of dispute data
End of alliance data
Louvain signed TIBI
Louvain trade
Louvain rescaled TIBI
IntraMax
AN

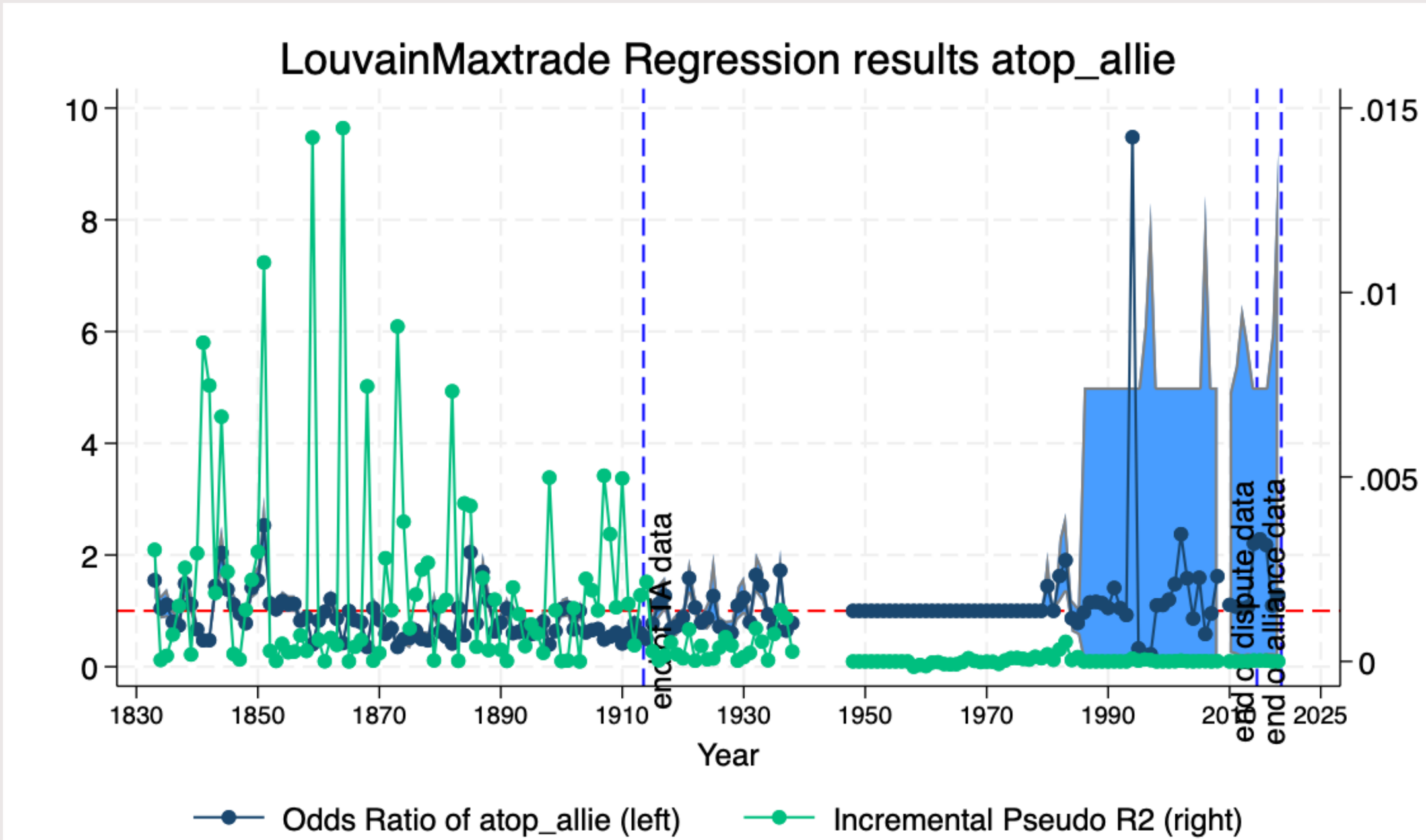

LouvainMaxtrade Regression results TA
Year
end of TA data
end of dispute data
end of alliance data
Odds Ratio of TA (left)
Incremental Pseudo R2 (right)

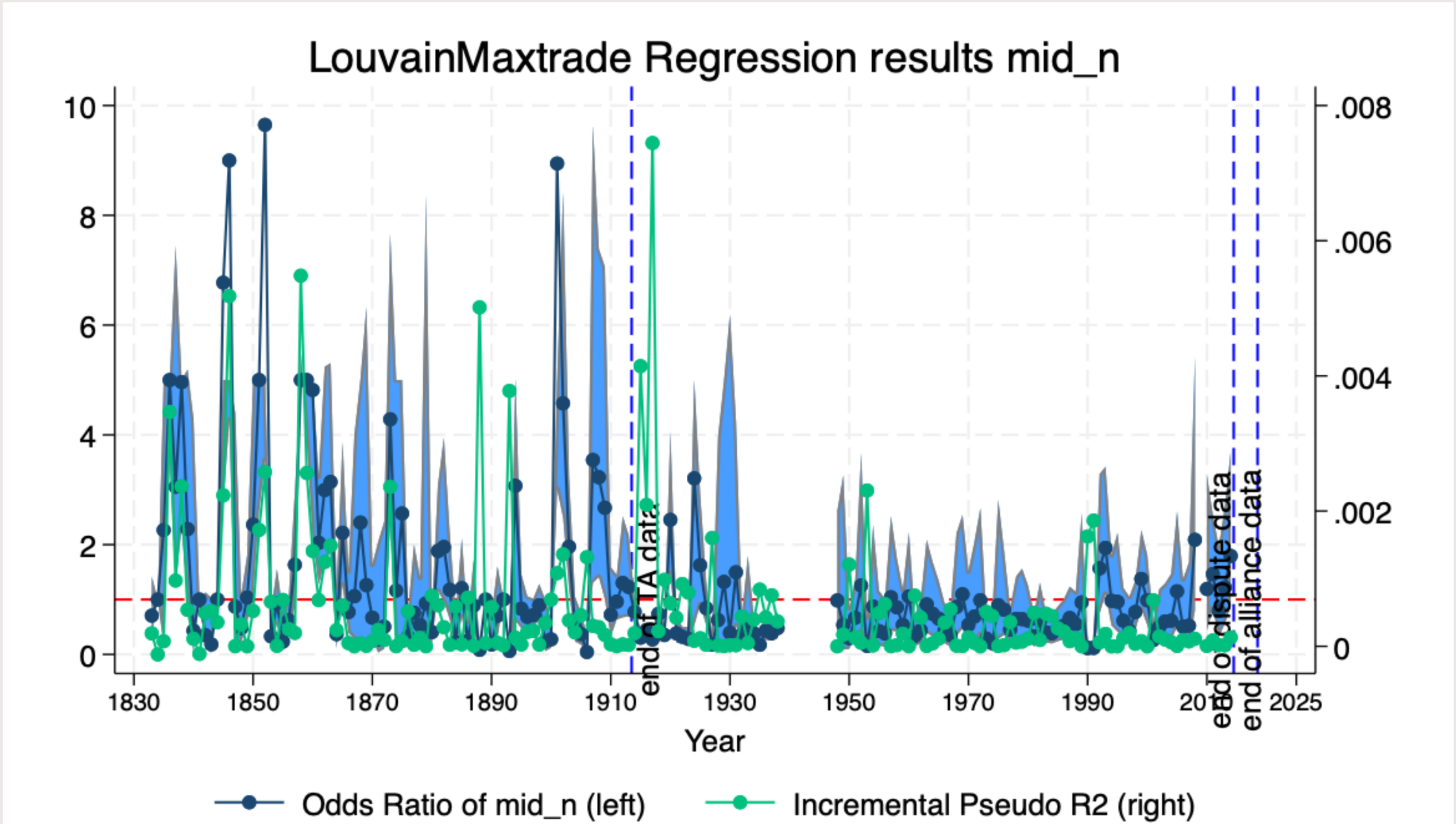

LouvainMaxtrade Regression results atop_allie
Year
end of TA data
end of dispute data
end of alliance data
Odds Ratio of atop_allie (left)
Incremental Pseudo R2 (right)

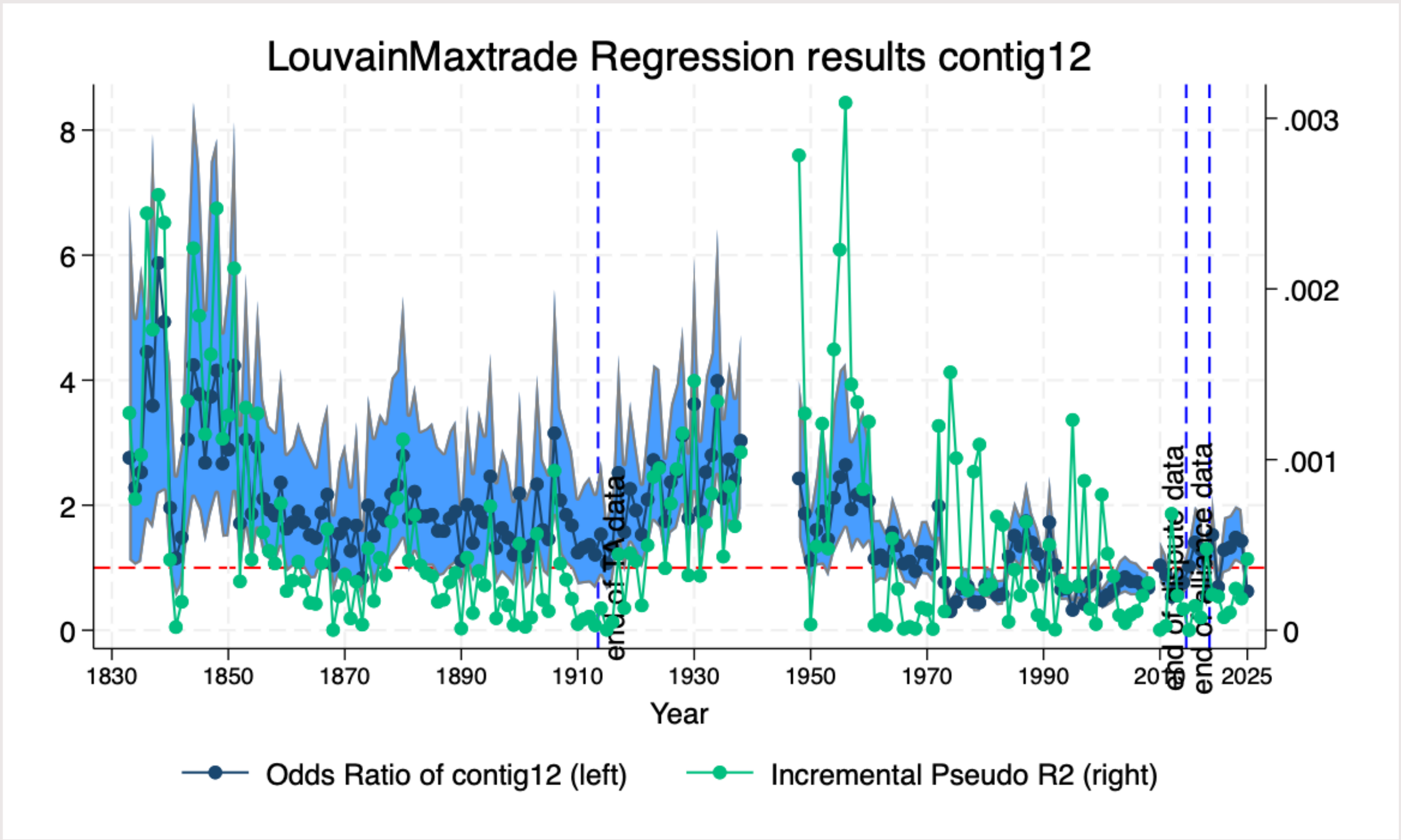

LouvainMaxtrade Regression results mid_n
Year
end of TA data
end of dispute data
end of alliance data
Odds Ratio of mid_n (left)
Incremental Pseudo R2 (right)

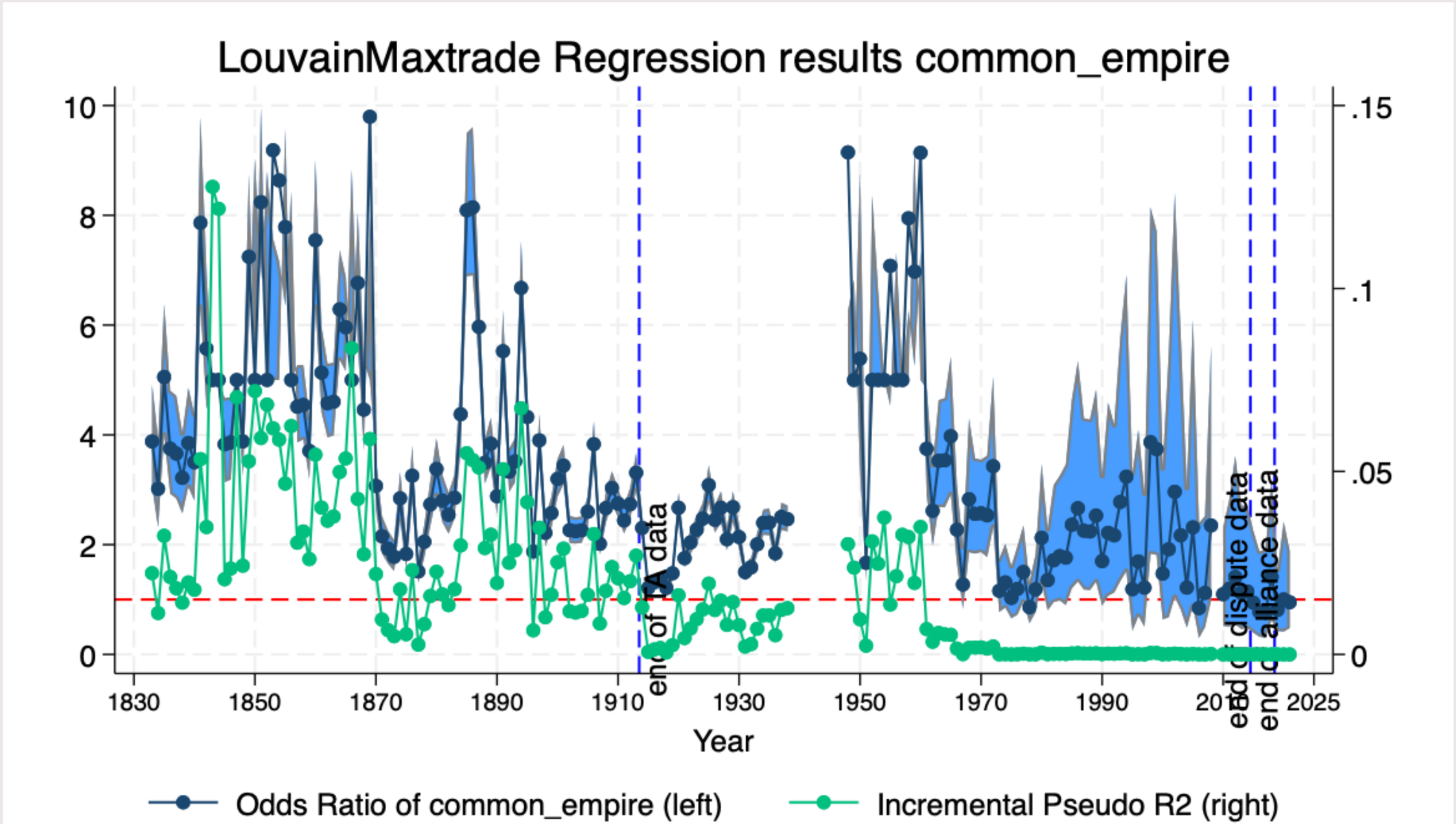

LouvainMaxtrade Regression results contig12
Year
end of TA data
end of dispute data
end of alliance data
Odds Ratio of contig12 (left)
Incremental Pseudo R2 (right)

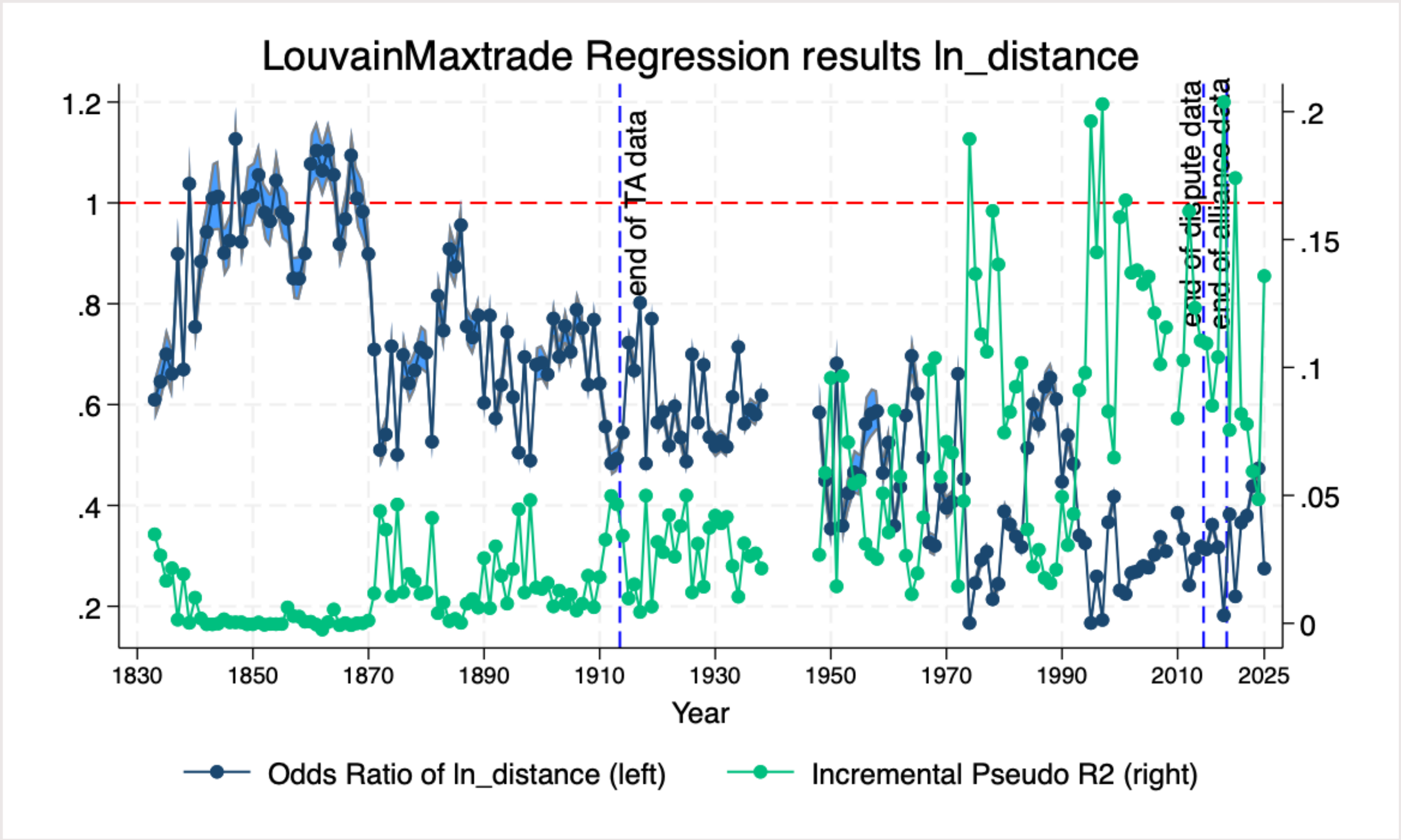

LouvainMaxtrade Regression results common_empire
Year
end of TA data
end of dispute data
end of alliance data
Odds Ratio of common_empire (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results atop_allie
10
8
6
4
2
0
.015
.01
.005
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
end of TA data
end of dispute data
end of alliance data
Year
Odds Ratio of atop_allie (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results mid_n
10
8
6
4
2
0
.008
.006
.004
.002
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
end of TA data
end of dispute data
end of alliance data
Year
Odds Ratio of mid_n (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results contig12
8
6
4
2
0
.003
.002
.001
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2025
Year
end of T4 data
end of dispute data
end of alliance data
Odds Ratio of contig12 (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results common_empire
10
8
6
4
2
0
.15
.1
.05
0
1830
1850
1870
1890
1910
1930
1950
1970
1990
2025
end of TA data
end of dispute data
end of alliance data
Year
Odds Ratio of common_empire (left)
Incremental Pseudo R2 (right)


LouvainMaxtrade Regression results ln_distance
1.2
1
.8
.6
.4
.2
.2
.15
.1
.05
0
end of TA data
end of dispute data
end of alliance data
1830
1850
1870
1890
1910
1930
1950
1970
1990
2010
2025
Year
Odds Ratio of ln_distance (left)
Incremental Pseudo R2 (right)

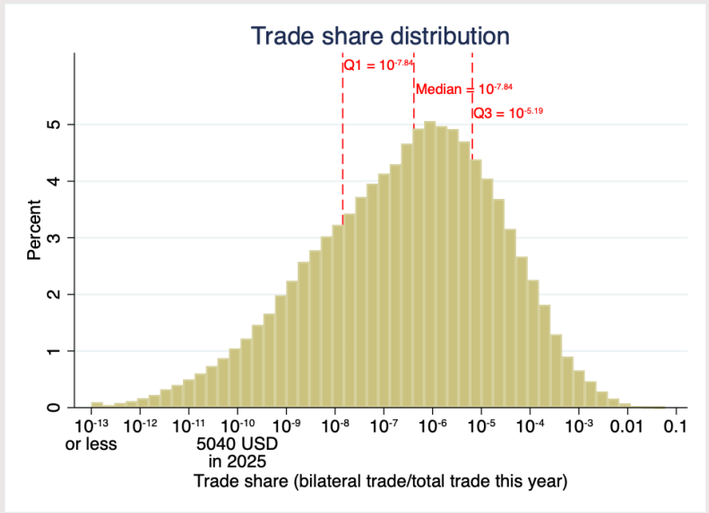
IV. EFFECTS OF TRADE COMMUNITIES

Effects of trade communities (1)

Importance of the extensive margin (Helpman, Melitz & Rubinstein (2008), Hummels & Klenow (2005) and Felbermayr & Kohler (2006))

Basic model in the trade duration literature: Cox model with proportional hazard (Besedeš and Prusa (2006))

Method : cloglog. Said to be preferable to Cox model in discrete time (Hess and Persson (2012)



We have an issue with the gravity model as it creates very small non-zero trade flows. So we probably want to cut off very small trade flows -- we choose one tenth of a billionth (c. 5k $ in 2025). Same question with failed splits

Trading flows

Excluding ↓ / Including very small flows and failed splits →	0	1	Total
0	7,867,646	158,949	8,026,595
1		2,613,141	2,613,141
Total	7,867,646	2,772,090	10,639,736

Trading pairs

Excluding ↓ / Including very small flows and failed splits →	0	1	Total
0	89,212	89,235	178,447
1		92,981	92,981
Total	89,212	182,216	271,428

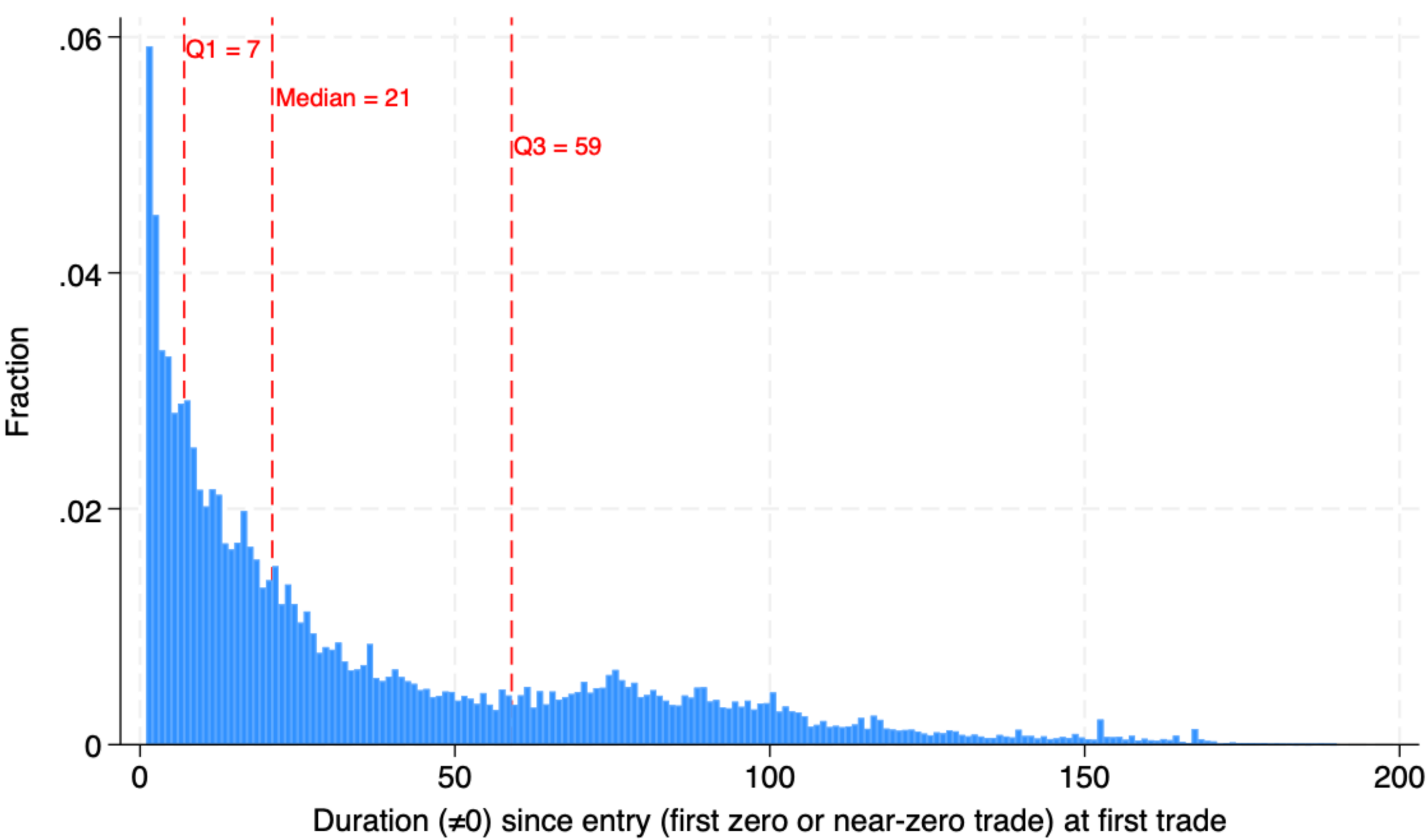
TRADING FLOWS "AT RISK" (EXCLUDING SECOND AND SUBSEQUENT TRADE OBSERVATIONS)

Excluding very small flows and failed splits	Total
0	6,387,135
1	86,158
Total	6,473,293

Including very small flows and failed splits	Total
0	6,220,999
1	91,083
Total	6,312,082

Time at risk

- Duration 1 : years of directed pair existence before first trade (not counting gaps)
- Duration 2 : time since appearance (counting gaps)
- Duration 3 : time since appearance with zero (and/or very small) trade


Q1 = 7
Median = 21
Q3 = 59
.06
.04
.02
0
Fraction
0
50
100
150
200
Duration (≠0) since entry (first zero or near-zero trade) at first trade

Community definition : LouvainMaxtrade (cloglog, year fixed effects) -- Hazard ratios

	Start trade (excluding very small flows and failed splits)			Start trade (including very small flows and failed splits)		
	Time since appearance with zero trade	Years of existence	Time since appearance	Time since appearance with zero trade	Years of existence	Time since appearance
	LouvainMaxtrade	LouvainMaxtrade	LouvainMaxtrade	LouvainMaxtrade	LouvainMaxtrade	LouvainMaxtrade
Duration (years)	0.9976***	0.9986***	0.9961***	0.9976***	0.9986***	0.9962***
	[0.9970, 0.9982]	[0.9979, 0.9994]	[0.9956, 0.9967]	[0.9970, 0.9982]	[0.9978, 0.9993]	[0.9956, 0.9967]
Duration²	1.000042***	1.000051***	1.000043***	1.000041***	1.000048***	1.000043***
	[1.000037, 1.000047]	[1.000045, 1.000057]	[1.000040, 1.000047]	[1.000037, 1.000046]	[1.000042, 1.000054]	[1.000039, 1.000047]
Nb. years in same community	0.99086***	0.98881***	0.99298***	0.99067***	0.98922***	0.99272***
	[0.99002, 0.99170]	[0.98777, 0.98986]	[0.99213, 0.99383]	[0.98984, 0.99149]	[0.98818, 0.99026]	[0.99188, 0.99356]
Share years in same community	1.33***	1.35***	1.23***	1.33***	1.33***	1.24***
	[1.29, 1.38]	[1.31, 1.39]	[1.20, 1.27]	[1.28, 1.37]	[1.29, 1.37]	[1.20, 1.27]
Ln(distance)	0.68***	0.67***	0.68***	0.69***	0.69***	0.69***
	[0.67, 0.69]	[0.66, 0.68]	[0.67, 0.68]	[0.68, 0.70]	[0.68, 0.70]	[0.68, 0.70]
Same empire	1.70***	1.85***	1.84***	1.76***	1.90***	1.89***
	[1.64, 1.77]	[1.79, 1.92]	[1.78, 1.91]	[1.69, 1.83]	[1.84, 1.96]	[1.82, 1.95]
Contiguity	1.44***	1.63***	1.64***	1.35***	1.54***	1.55***
	[1.26, 1.66]	[1.45, 1.82]	[1.47, 1.84]	[1.17, 1.56]	[1.36, 1.73]	[1.38, 1.75]
Conflict (MID)	0.52***	0.58***	0.59***	0.57***	0.65***	0.66***
	[0.42, 0.64]	[0.49, 0.69]	[0.49, 0.70]	[0.47, 0.70]	[0.55, 0.77]	[0.56, 0.78]
Alliance (ATOP)	3.35***	3.91***	3.93***	2.65***	3.08***	3.09***
	[3.01, 3.73]	[3.59, 4.27]	[3.60, 4.29]	[2.41, 2.91]	[2.86, 3.32]	[2.87, 3.34]
Trade agreement	3.27***	3.87***	3.90***	3.13***	3.75***	3.78***
	[2.70, 3.95]	[3.34, 4.47]	[3.37, 4.52]	[2.60, 3.78]	[3.24, 4.33]	[3.27, 4.37]
Observations	4,147,820	6,190,724	6,190,724	3,992,075	6,030,436	6,030,436
Pairs	161,646	240,905	240,905	160,402	240,117	240,117
Log-likelihood	-360752.8	-412705.3	-412667.2	-371911.1	-427428.3	-427334.1

*** p<.01, ** p<.05, * p<.1 -- 95 % confidence intervals in brackets.

	Start trade (excluding very small flows and failed splits)				
	Time since appearance with zero trade				
	LouvainMaxtrade	IntraMax	Tibi	LouvainTibirescale	AN
Duration (years)	0.9976***	0.9980***	0.9970***	0.9969***	0.9957***
	[0.9970, 0.9982]	[0.9974, 0.9987]	[0.9963, 0.9976]	[0.9963, 0.9976]	[0.9951, 0.9963]
Duration²	1.000042***	1.000040***	1.000044***	1.000043***	1.000046***
	[1.000037, 1.000047]	[1.000035, 1.000045]	[1.000039, 1.000049]	[1.000038, 1.000048]	[1.000041, 1.000051]
Nb. years in same community	0.99086***	0.99048***	0.99120***	0.99147***	0.99923**
	[0.99002, 0.99170]	[0.98957, 0.99139]	[0.99034, 0.99206]	[0.99047, 0.99246]	[0.99856, 0.99989]
Share years in same community	1.33***	1.49***	1.15***	1.30***	0.81***
	[1.29, 1.38]	[1.43, 1.54]	[1.10, 1.19]	[1.24, 1.35]	[0.78, 0.84]
Ln(distance)	0.68***	0.68***	0.67***	0.67***	0.63***
	[0.67, 0.69]	[0.67, 0.69]	[0.66, 0.68]	[0.66, 0.69]	[0.61, 0.64]
Same empire	1.70***	1.69***	1.70***	1.71***	1.73***
	[1.64, 1.77]	[1.63, 1.76]	[1.64, 1.77]	[1.64, 1.78]	[1.66, 1.81]
Contiguity	1.44***	1.45***	1.44***	1.43***	1.46***
	[1.26, 1.66]	[1.27, 1.66]	[1.25, 1.66]	[1.24, 1.64]	[1.26, 1.70]
Conflict (MID)	0.52***	0.52***	0.52***	0.52***	0.52***
	[0.42, 0.64]	[0.43, 0.65]	[0.42, 0.64]	[0.42, 0.64]	[0.42, 0.64]
Alliance (ATOP)	3.35***	3.29***	3.35***	3.36***	3.32***
	[3.01, 3.73]	[2.96, 3.66]	[3.01, 3.73]	[3.02, 3.74]	[2.98, 3.70]
Trade agreement	3.27***	3.22***	3.24***	3.25***	3.05***
	[2.70, 3.95]	[2.66, 3.90]	[2.68, 3.92]	[2.69, 3.93]	[2.50, 3.72]
Observations	4,147,820	4,147,820	4,147,820	4,147,820	4,147,820
Pairs	161,646	161,646	161,646	161,646	161,646
Log-likelihood	-360752.8	-360692.3	-360788.2	-360871.9	-360797.4

*** p<.01, ** p<.05, * p<.1 95 % confidence intervals in brackets.
dur3 = dur_n_fs for flag_trade_entry_n_fs

CONCLUSION

Outline

1. Data
2. Identifying trade communities
3. Finding the determinants of trade communities
4. Finding the effect of trade communities

Tentative findings

1. Chronology of the importance of trade communities
2. They are not pure artefacts

Pat on the back

Much yet to be done

- RICardo: [https://ricardo.medialab.sciences-po.fr]
- GeoPolHist: https://medialab.github.io/GeoPolHist